BCB546 R Assignment

Prince Ameyaw

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```r
# --- Data Acquisition ---
# Note: This script uses relative paths.
# Ensure the .txt files are in the same directory as this Rmd file.
```

```r
## Load library
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(dplyr)
```

```r
# Load data

## load fang_et_al_genotypes.txt file
genoty_data <- read_tsv("fang_et_al_genotypes.txt")
```

```
## Rows: 2782 Columns: 986
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (986): Sample_ID, JG_OTU, Group, abph1.20, abph1.22, ae1.3, ae1.4, ae1.5...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
file.size("fang_et_al_genotypes.txt")
```

```
## [1] 11051939
```

```r
## load snp_position.txt file
snp_pos_data <- read_tsv("snp_position.txt")
```

```
## Rows: 983 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (9): SNP_ID, Chromosome, Position, alt_pos, mult_positions, amplicon, cd...
## dbl (6): cdv_marker_id, Genaissance_daa_id, Sequenom_daa_id, count_amplicons...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
file.size("snp_position.txt")
```

```
## [1] 82763
```

```r
# data inspection; Verify data integrity, dimensions, and structure before processing

## fang_et_al_genotypes.txt
str(genoty_data)
```

```
## spc_tbl_ [2,782 x 986] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Sample_ID     : chr [1:2782] "SL-15" "SL-16" "SL-11" "SL-12" ...
##  $ JG_OTU        : chr [1:2782] "T-aust-1" "T-aust-2" "T-brav-1" "T-brav-2" ...
##  $ Group         : chr [1:2782] "TRIPS" "TRIPS" "TRIPS" "TRIPS" ...
##  $ abph1.20      : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ abph1.22      : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ ae1.3         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ ae1.4         : chr [1:2782] "G/G" "?/?" "G/G" "G/G" ...
##  $ ae1.5         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ an1.4         : chr [1:2782] "C/C" "C/C" "?/?" "C/C" ...
##  $ ba1.6         : chr [1:2782] "?/?" "A/G" "G/G" "G/G" ...
##  $ ba1.9         : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ bt2.5         : chr [1:2782] "?/?" "?/?" "C/C" "C/C" ...
##  $ bt2.7         : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ bt2.8         : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ Fea2.1        : chr [1:2782] "C/C" "C/C" "?/?" "?/?" ...
##  $ Fea2.5        : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ id1.3         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ lg2.11        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ lg2.2         : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ pbf1.1        : chr [1:2782] "?/?" "T/T" "T/T" "T/T" ...
##  $ pbf1.2        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.3        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.5        : chr [1:2782] "?/?" "?/?" "A/A" "A/A" ...
##  $ pbf1.6        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.7        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ pbf1.8        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00003.11   : chr [1:2782] "?/?" "?/?" "C/C" "?/?" ...
##  $ PZA00004.2    : chr [1:2782] "T/T" "T/T" "?/?" "T/T" ...
##  $ PZA00005.8    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00005.9    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
```

```
##  $ PZA00006.13   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00006.14   : chr [1:2782] "?/?" "G/G" "G/G" "G/G" ...
##  $ PZA00008.1    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00010.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00013.10   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00013.11   : chr [1:2782] "C/C" "C/C" "C/T" "C/T" ...
##  $ PZA00013.9    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00015.4    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00017.1    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00018.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00029.11   : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00029.12   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00030.11   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00031.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00041.3    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00042.2    : chr [1:2782] "?/?" "T/T" "?/?" "?/?" ...
##  $ PZA00042.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00043.7    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00045.1    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00047.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00049.12   : chr [1:2782] "?/?" "?/?" "T/T" "T/T" ...
##  $ PZA00050.9    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00051.2    : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00058.5    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00058.6    : chr [1:2782] "T/T" "?/?" "T/T" "T/T" ...
##  $ PZA00060.2    : chr [1:2782] "?/?" "?/?" "C/C" "C/C" ...
##  $ PZA00061.1    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00065.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00069.4    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00070.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00078.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00079.1    : chr [1:2782] "C/C" "?/?" "C/C" "C/C" ...
##  $ PZA00081.17   : chr [1:2782] "?/?" "T/T" "T/T" "T/T" ...
##  $ PZA00084.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00084.3    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00086.8    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00088.3    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00090.2    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00092.1    : chr [1:2782] "?/?" "?/?" "T/T" "?/?" ...
##  $ PZA00092.5    : chr [1:2782] "?/?" "C/C" "C/C" "?/?" ...
##  $ PZA00093.2    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00096.26   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00097.13   : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00098.14   : chr [1:2782] "?/?" "?/?" "C/C" "?/?" ...
##  $ PZA00100.10   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00100.12   : chr [1:2782] "T/T" "?/?" "T/T" "T/T" ...
##  $ PZA00100.14   : chr [1:2782] "?/?" "A/A" "A/A" "A/A" ...
##  $ PZA00100.9    : chr [1:2782] "C/C" "C/C" "C/C" "?/?" ...
##  $ PZA00103.20   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00106.9    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00107.18   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00108.12   : chr [1:2782] "?/?" "C/C" "C/C" "C/C" ...
##  $ PZA00108.14   : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00108.15   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
```

```
##  $ PZA00109.3    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00109.5    : chr [1:2782] "A/A" "?/?" "A/A" "?/?" ...
##  $ PZA00111.2    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00111.4    : chr [1:2782] "C/C" "?/?" "C/C" "C/C" ...
##  $ PZA00111.5    : chr [1:2782] "?/?" "?/?" "A/A" "A/A" ...
##  $ PZA00111.6    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00111.8    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00114.3    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00116.2    : chr [1:2782] "C/T" "C/T" "C/T" "C/T" ...
##  $ PZA00119.4    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00120.4    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00123.1    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00125.2    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00131.14   : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00132.17   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##   [list output truncated]
##  - attr(*, "spec")=
##   .. cols(
##   ..   Sample_ID = col_character(),
##   ..   JG_OTU = col_character(),
##   ..   Group = col_character(),
##   ..   abph1.20 = col_character(),
##   ..   abph1.22 = col_character(),
##   ..   ae1.3 = col_character(),
##   ..   ae1.4 = col_character(),
##   ..   ae1.5 = col_character(),
##   ..   an1.4 = col_character(),
##   ..   ba1.6 = col_character(),
##   ..   ba1.9 = col_character(),
##   ..   bt2.5 = col_character(),
##   ..   bt2.7 = col_character(),
##   ..   bt2.8 = col_character(),
##   ..   Fea2.1 = col_character(),
##   ..   Fea2.5 = col_character(),
##   ..   id1.3 = col_character(),
##   ..   lg2.11 = col_character(),
##   ..   lg2.2 = col_character(),
##   ..   pbf1.1 = col_character(),
##   ..   pbf1.2 = col_character(),
##   ..   pbf1.3 = col_character(),
##   ..   pbf1.5 = col_character(),
##   ..   pbf1.6 = col_character(),
##   ..   pbf1.7 = col_character(),
##   ..   pbf1.8 = col_character(),
##   ..   PZA00003.11 = col_character(),
##   ..   PZA00004.2 = col_character(),
##   ..   PZA00005.8 = col_character(),
##   ..   PZA00005.9 = col_character(),
##   ..   PZA00006.13 = col_character(),
##   ..   PZA00006.14 = col_character(),
##   ..   PZA00008.1 = col_character(),
##   ..   PZA00010.5 = col_character(),
##   ..   PZA00013.10 = col_character(),
##   ..   PZA00013.11 = col_character(),
```

```
##   ..   PZA00013.9 = col_character(),
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##   ..   PZA00030.11 = col_character(),
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##   ..   PZA00060.2 = col_character(),
##   ..   PZA00061.1 = col_character(),
##   ..   PZA00065.2 = col_character(),
##   ..   PZA00069.4 = col_character(),
##   ..   PZA00070.5 = col_character(),
##   ..   PZA00078.2 = col_character(),
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##   ..   PZA00081.17 = col_character(),
##   ..   PZA00084.2 = col_character(),
##   ..   PZA00084.3 = col_character(),
##   ..   PZA00086.8 = col_character(),
##   ..   PZA00088.3 = col_character(),
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##   ..   PZA00103.20 = col_character(),
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##   ..   PZA00111.6 = col_character(),
```

```
##   ..   PZA00111.8 = col_character(),
##   ..   PZA00114.3 = col_character(),
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##   ..   PZA00119.4 = col_character(),
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##   ..   PZA00135.6 = col_character(),
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##   ..   PZA00142.6 = col_character(),
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##   ..   PZA00166.1 = col_character(),
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##   ..   PZA00170.1 = col_character(),
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##   ..   PZA00170.4 = col_character(),
##   ..   PZA00174.1 = col_character(),
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##   ..   PZA00178.3 = col_character(),
##   ..   PZA00182.3 = col_character(),
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##   ..   PZA00184.1 = col_character(),
##   ..   PZA00184.4 = col_character(),
##   ..   PZA00188.1 = col_character(),
##   ..   PZA00188.3 = col_character(),
##   ..   PZA00191.5 = col_character(),
##   ..   PZA00192.6 = col_character(),
##   ..   PZA00192.7 = col_character(),
##   ..   PZA00193.2 = col_character(),
##   ..   PZA00198.39 = col_character(),
##   ..   PZA00200.11 = col_character(),
##   ..   PZA00200.17 = col_character(),
##   ..   PZA00200.9 = col_character(),
##   ..   PZA00201.2 = col_character(),
##   ..   PZA00204.1 = col_character(),
##   ..   PZA00210.1 = col_character(),
##   ..   PZA00210.6 = col_character(),
```

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##   ..   PZA00211.7 = col_character(),
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##   ..   PZA00216.9 = col_character(),
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##   ..   PZA00218.6 = col_character(),
##   ..   PZA00219.7 = col_character(),
##   ..   PZA00220.11 = col_character(),
##   ..   PZA00220.12 = col_character(),
##   ..   PZA00221.7 = col_character(),
##   ..   PZA00225.8 = col_character(),
##   ..   PZA00226.7 = col_character(),
##   ..   PZA00227.8 = col_character(),
##   ..   PZA00230.5 = col_character(),
##   ..   PZA00232.24 = col_character(),
##   ..   PZA00234.21 = col_character(),
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##   ..   PZA00237.8 = col_character(),
##   ..   PZA00238.3 = col_character(),
##   ..   PZA00240.9 = col_character(),
##   ..   PZA00241.6 = col_character(),
##   ..   PZA00243.27 = col_character(),
##   ..   PZA00245.14 = col_character(),
##   ..   PZA00245.16 = col_character(),
##   ..   PZA00245.17 = col_character(),
##   ..   PZA00245.18 = col_character(),
##   ..   PZA00245.19 = col_character(),
##   ..   PZA00249.2 = col_character(),
##   ..   PZA00250.1 = col_character(),
##   ..   PZA00251.1 = col_character(),
##   ..   PZA00254.3 = col_character(),
##   ..   PZA00255.15 = col_character(),
##   ..   PZA00255.17 = col_character(),
##   ..   PZA00256.16 = col_character(),
##   ..   PZA00256.21 = col_character(),
##   ..   PZA00256.23 = col_character(),
##   ..   PZA00257.11 = col_character(),
##   ..   PZA00257.22 = col_character(),
##   ..   PZA00261.6 = col_character(),
##   ..   PZA00263.14 = col_character(),
##   ..   PZA00266.5 = col_character(),
##   ..   PZA00270.3 = col_character(),
##   ..   PZA00273.1 = col_character(),
##   ..   PZA00274.7 = col_character(),
##   ..   PZA00277.17 = col_character(),
##   ..   PZA00277.9 = col_character(),
##   ..   PZA00280.14 = col_character(),
##   ..   PZA00287.1 = col_character(),
##   ..   PZA00289.11 = col_character(),
##   ..   PZA00294.20 = col_character(),
```

```
##   ..   PZA00296.6 = col_character(),
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##   ..   PZA00364.6 = col_character(),
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##   ..   PZA00369.1 = col_character(),
##   ..   PZA00370.1 = col_character(),
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##   ..   PZA03283.2 = col_character(),
##   ..   PZA03284.3 = col_character(),
##   ..   PZA03290.1 = col_character(),
##   ..   PZA03290.2 = col_character(),
##   ..   PZA03295.4 = col_character(),
##   ..   PZA03296.6 = col_character(),
##   ..   PZA03296.7 = col_character(),
##   ..   PZA03298.1 = col_character(),
##   ..   PZA03298.2 = col_character(),
##   ..   PZA03301.2 = col_character(),
```

```
##   ..   PZA03301.4 = col_character(),
##   ..   PZA03302.1 = col_character(),
##   ..   PZA03305.6 = col_character(),
##   ..   PZA03305.7 = col_character(),
##   ..   PZA03311.2 = col_character(),
##   ..   PZA03311.3 = col_character(),
##   ..   PZA03311.4 = col_character(),
##   ..   PZA03311.5 = col_character(),
##   ..   PZA03312.1 = col_character(),
##   ..   PZA03312.2 = col_character(),
##   ..   PZA03316.2 = col_character(),
##   ..   PZA03317.1 = col_character(),
##   ..   PZA03319.3 = col_character(),
##   ..   PZA03319.4 = col_character(),
##   ..   PZA03320.3 = col_character(),
##   ..   PZA03320.4 = col_character(),
##   ..   PZA03328.5 = col_character(),
##   ..   PZA03329.1 = col_character(),
##   ..   PZA03329.2 = col_character(),
##   ..   PZA03333.3 = col_character(),
##   ..   PZA03335.2 = col_character(),
##   ..   PZA03335.3 = col_character(),
##   ..   PZA03337.1 = col_character(),
##   ..   PZA03338.5 = col_character(),
##   ..   PZA03340.2 = col_character(),
##   ..   PZA03342.2 = col_character(),
##   ..   PZA03344.4 = col_character(),
##   ..   PZA03344.5 = col_character(),
##   ..   PZA03344.6 = col_character(),
##   ..   PZA03345.1 = col_character(),
##   ..   PZA03345.2 = col_character(),
##   ..   PZA03345.4 = col_character(),
##   ..   PZA03347.1 = col_character(),
##   ..   PZA03348.1 = col_character(),
##   ..   PZA03349.1 = col_character(),
##   ..   PZA03349.9 = col_character(),
##   ..   PZA03767.1 = col_character(),
##   ..   PZA03767.4 = col_character(),
##   ..   PZA03767.5 = col_character(),
##   ..   PZA03773.2 = col_character(),
##   ..   PZA03773.3 = col_character(),
##   ..   PZA03774.1 = col_character(),
##   ..   PZA03774.10 = col_character(),
##   ..   PZA03774.2 = col_character(),
##   ..   PZA03774.4 = col_character(),
##   ..   PZA03774.5 = col_character(),
##   ..   PZA03774.6 = col_character(),
##   ..   PZA03774.8 = col_character(),
##   ..   PZA03774.9 = col_character(),
##   ..   PZA03775.1 = col_character(),
##   ..   PZA03775.11 = col_character(),
##   ..   PZA03775.2 = col_character(),
##   ..   PZA03775.3 = col_character(),
##   ..   PZA03775.4 = col_character(),
```

```
##   ..   PZA03775.6 = col_character(),
##   ..   PZA03775.7 = col_character(),
##   ..   PZA03775.8 = col_character(),
##   ..   PZA03775.9 = col_character(),
##   ..   PZA03781.1 = col_character(),
##   ..   PZA03781.2 = col_character(),
##   ..   PZA03781.3 = col_character(),
##   ..   PZA03781.4 = col_character(),
##   ..   PZA03781.5 = col_character(),
##   ..   PZA03781.6 = col_character(),
##   ..   PZA03781.7 = col_character(),
##   ..   PZA03781.8 = col_character(),
##   ..   PZA03782.1 = col_character(),
##   ..   PZA03782.3 = col_character(),
##   ..   PZA03786.1 = col_character(),
##   ..   PZA03786.2 = col_character(),
##   ..   PZA03789.1 = col_character(),
##   ..   PZA03789.2 = col_character(),
##   ..   PZA03789.4 = col_character(),
##   ..   PZB00011.4 = col_character(),
##   ..   PZB00011.5 = col_character(),
##   ..   PZB00041.2 = col_character(),
##   ..   PZB00041.4 = col_character(),
##   ..   PZB00049.2 = col_character(),
##   ..   PZB00049.4 = col_character(),
##   ..   PZB00049.7 = col_character(),
##   ..   PZB00055.1 = col_character(),
##   ..   PZB00060.4 = col_character(),
##   ..   PZB00062.6 = col_character(),
##   ..   PZB00062.7 = col_character(),
##   ..   PZB00062.8 = col_character(),
##   ..   PZB00067.2 = col_character(),
##   ..   PZB00067.3 = col_character(),
##   ..   PZB00067.4 = col_character(),
##   ..   PZB00067.5 = col_character(),
##   ..   PZB00078.1 = col_character(),
##   ..   PZB00081.2 = col_character(),
##   ..   PZB00081.4 = col_character(),
##   ..   PZB00081.5 = col_character(),
##   ..   PZB00081.7 = col_character(),
##   ..   PZB00092.1 = col_character(),
##   ..   PZB00092.4 = col_character(),
##   ..   PZB00093.3 = col_character(),
##   ..   PZB00093.4 = col_character(),
##   ..   PZB00093.6 = col_character(),
##   ..   PZB00096.2 = col_character(),
##   ..   PZB00096.3 = col_character(),
##   ..   PZB00136.3 = col_character(),
##   ..   PZB00140.1 = col_character(),
##   ..   PZB00145.2 = col_character(),
##   ..   PZB00149.2 = col_character(),
##   ..   PZB00149.4 = col_character(),
##   ..   PZB00153.1 = col_character(),
##   ..   PZB00153.2 = col_character(),
```

```
##   ..   PZB00153.3 = col_character(),
##   ..   PZB00153.5 = col_character(),
##   ..   PZB00160.1 = col_character(),
##   ..   PZB00160.2 = col_character(),
##   ..   PZB00160.4 = col_character(),
##   ..   PZB00165.2 = col_character(),
##   ..   PZB00165.6 = col_character(),
##   ..   PZB00169.4 = col_character(),
##   ..   PZB00169.6 = col_character(),
##   ..   PZB00175.1 = col_character(),
##   ..   PZB00175.2 = col_character(),
##   ..   PZB00175.3 = col_character(),
##   ..   PZB00175.4 = col_character(),
##   ..   PZB00175.5 = col_character(),
##   ..   PZB00180.1 = col_character(),
##   ..   PZB00180.2 = col_character(),
##   ..   PZB00183.3 = col_character(),
##   ..   PZB00188.6 = col_character(),
##   ..   PZB00207.3 = col_character(),
##   ..   PZB00221.3 = col_character(),
##   ..   PZB00221.8 = col_character(),
##   ..   PZB00229.3 = col_character(),
##   ..   PZB00232.1 = col_character(),
##   ..   PZB00232.2 = col_character(),
##   ..   PZB00232.4 = col_character(),
##   ..   PZB00232.5 = col_character(),
##   ..   PZB00379.3 = col_character(),
##   ..   PZB00379.4 = col_character(),
##   ..   PZB00379.5 = col_character(),
##   ..   PZB00393.7 = col_character(),
##   ..   PZB00409.3 = col_character(),
##   ..   PZB00416.2 = col_character(),
##   ..   PZB00416.5 = col_character(),
##   ..   PZB00454.2 = col_character(),
##   ..   PZB00454.3 = col_character(),
##   ..   PZB00454.4 = col_character(),
##   ..   PZB00454.5 = col_character(),
##   ..   PZB00498.2 = col_character(),
##   ..   PZB00498.4 = col_character(),
##   ..   PZB00598.1 = col_character(),
##   ..   PZB00598.2 = col_character(),
##   ..   PZB00603.3 = col_character(),
##   ..   PZB00603.4 = col_character(),
##   ..   PZB00603.5 = col_character(),
##   ..   PZB00607.2 = col_character(),
##   ..   PZB00761.1 = col_character(),
##   ..   PZB00761.2 = col_character(),
##   ..   PZB00849.2 = col_character(),
##   ..   PZB00849.3 = col_character(),
##   ..   PZB00849.4 = col_character(),
##   ..   PZB00859.1 = col_character(),
##   ..   PZB01109.2 = col_character(),
##   ..   PZB01109.3 = col_character(),
##   ..   PZB01110.1 = col_character(),
```

```
##   ..   PZB01110.2 = col_character(),
##   ..   PZB01110.3 = col_character(),
##   ..   PZB01111.6 = col_character(),
##   ..   PZB01111.7 = col_character(),
##   ..   PZB01111.8 = col_character(),
##   ..   PZB01112.3 = col_character(),
##   ..   PZB01112.4 = col_character(),
##   ..   PZB01112.5 = col_character(),
##   ..   PZB01112.6 = col_character(),
##   ..   PZB01113.4 = col_character(),
##   ..   PZB01114.1 = col_character(),
##   ..   PZB01114.3 = col_character(),
##   ..   PZB01115.1 = col_character(),
##   ..   PZB01115.5 = col_character(),
##   ..   PZB01115.6 = col_character(),
##   ..   PZB01116.2 = col_character(),
##   ..   PZB01221.1 = col_character(),
##   ..   PZB01222.1 = col_character(),
##   ..   PZB01222.3 = col_character(),
##   ..   PZB01223.3 = col_character(),
##   ..   PZB01223.4 = col_character(),
##   ..   PZB01223.7 = col_character(),
##   ..   PZB01225.1 = col_character(),
##   ..   PZB01225.2 = col_character(),
##   ..   PZB01225.4 = col_character(),
##   ..   PZB01228.1 = col_character(),
##   ..   PZB01228.3 = col_character(),
##   ..   PZB01228.4 = col_character(),
##   ..   PZB01233.2 = col_character(),
##   ..   PZB01233.3 = col_character(),
##   ..   PZB01238.5 = col_character(),
##   ..   PZB01238.6 = col_character(),
##   ..   PZB01427.1 = col_character(),
##   ..   PZB01427.3 = col_character(),
##   ..   PZB01463.2 = col_character(),
##   ..   PZB01463.3 = col_character(),
##   ..   PZB01463.4 = col_character(),
##   ..   PZD00003.1 = col_character(),
##   ..   PZD00003.3 = col_character(),
##   ..   PZD00007.1 = col_character(),
##   ..   PZD00008.3 = col_character(),
##   ..   PZD00011.1 = col_character(),
##   ..   PZD00011.3 = col_character(),
##   ..   PZD00011.4 = col_character(),
##   ..   PZD00012.1 = col_character(),
##   ..   PZD00012.2 = col_character(),
##   ..   PZD00012.3 = col_character(),
##   ..   PZD00012.4 = col_character(),
##   ..   PZD00012.5 = col_character(),
##   ..   PZD00013.3 = col_character(),
##   ..   PZD00013.4 = col_character(),
##   ..   PZD00014.3 = col_character(),
##   ..   PZD00017.1 = col_character(),
##   ..   PZD00019.1 = col_character(),
```

```
##   ..   PZD00020.2 = col_character(),
##   ..   PZD00020.3 = col_character(),
##   ..   PZD00020.4 = col_character(),
##   ..   PZD00020.6 = col_character(),
##   ..   PZD00021.2 = col_character(),
##   ..   PZD00021.4 = col_character(),
##   ..   PZD00021.5 = col_character(),
##   ..   PZD00022.1 = col_character(),
##   ..   PZD00022.3 = col_character(),
##   ..   PZD00022.4 = col_character(),
##   ..   PZD00024.2 = col_character(),
##   ..   PZD00025.1 = col_character(),
##   ..   PZD00025.2 = col_character(),
##   ..   PZD00030.1 = col_character(),
##   ..   PZD00030.4 = col_character(),
##   ..   PZD00030.5 = col_character(),
##   ..   PZD00030.6 = col_character(),
##   ..   PZD00034.3 = col_character(),
##   ..   PZD00043.1 = col_character(),
##   ..   PZD00043.2 = col_character(),
##   ..   PZD00043.3 = col_character(),
##   ..   PZD00043.4 = col_character(),
##   ..   PZD00044.2 = col_character(),
##   ..   PZD00044.3 = col_character(),
##   ..   PZD00044.4 = col_character(),
##   ..   PZD00045.1 = col_character(),
##   ..   PZD00045.2 = col_character(),
##   ..   PZD00045.3 = col_character(),
##   ..   PZD00045.4 = col_character(),
##   ..   PZD00049.3 = col_character(),
##   ..   PZD00049.4 = col_character(),
##   ..   PZD00049.5 = col_character(),
##   ..   PZD00051.1 = col_character(),
##   ..   PZD00052.3 = col_character(),
##   ..   PZD00052.4 = col_character(),
##   ..   PZD00062.2 = col_character(),
##   ..   PZD00066.1 = col_character(),
##   ..   PZD00067.1 = col_character(),
##   ..   PZD00067.2 = col_character(),
##   ..   PZD00067.3 = col_character(),
##   ..   PZD00068.1 = col_character(),
##   ..   PZD00069.2 = col_character(),
##   ..   PZD00069.3 = col_character(),
##   ..   PZD00069.4 = col_character(),
##   ..   PZD00069.5 = col_character(),
##   ..   PZD00073.1 = col_character(),
##   ..   PZD00073.2 = col_character(),
##   ..   PZD00073.6 = col_character(),
##   ..   PZD00074.1 = col_character(),
##   ..   PZD00075.1 = col_character(),
##   ..   PZD00075.2 = col_character(),
##   ..   PZD00076.1 = col_character(),
##   ..   PZD00076.2 = col_character(),
##   ..   PZD00076.4 = col_character(),
```

```
##   ..   PZD00077.10 = col_character(),
##   ..   PZD00077.5 = col_character(),
##   ..   PZD00077.7 = col_character(),
##   ..   PZD00077.8 = col_character(),
##   ..   PZD00078.2 = col_character(),
##   ..   Ra2_ORF.1 = col_character(),
##   ..   Ra2_ORF.2 = col_character(),
##   ..   Ra2_ORF.4 = col_character(),
##   ..   Ra2_promoter.1 = col_character(),
##   ..   Ra2_promoter.2 = col_character(),
##   ..   Ra2_promoter.3 = col_character(),
##   ..   sh2.5 = col_character(),
##   ..   sh2.6 = col_character(),
##   ..   sh2.7 = col_character(),
##   ..   sh2.9 = col_character(),
##   ..   su1.4 = col_character(),
##   ..   su1.5 = col_character(),
##   ..   su1.7 = col_character(),
##   ..   tb1.17 = col_character(),
##   ..   tb1.18 = col_character(),
##   ..   tb1.19 = col_character(),
##   ..   tb1.5 = col_character(),
##   ..   te1.3 = col_character(),
##   ..   te1.4 = col_character(),
##   ..   zagl1.1 = col_character(),
##   ..   zagl1.6 = col_character(),
##   ..   zap1.2 = col_character(),
##   ..   zen1.1 = col_character(),
##   ..   zen1.2 = col_character(),
##   ..   zen1.4 = col_character(),
##   ..   zfl2.6 = col_character(),
##   ..   zmm3.4 = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
dim(genoty_data)
```

```
## [1] 2782  986
```

```r
head(genoty_data)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SL-15     T-aust-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 2 SL-16     T-aust-2 TRIPS ?/?      ?/?      T/T   ?/?   T/T   C/C   A/G   G/G
## 3 SL-11     T-brav-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   ?/?   G/G   G/G
## 4 SL-12     T-brav-2 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   G/G   G/G
## 5 SL-18     T-cund   TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 6 SL-2      T-dact-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   A/G   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
```

```
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
tail(genoty_data)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SYN262    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 2 S0398     Zmm-IL-~ ZMMIL G/G      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 3 S1636     Zmm-IL-~ ZMMIL G/G      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 4 CU0201    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 5 S0215     Zmm-IL-~ ZMMIL G/G      A/A      T/T   ?/?   C/C   C/C   G/G   G/G
## 6 CU0202    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   ?/?   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
## snp_position.txt
str(snp_pos_data)
```

```
## spc_tbl_ [983 x 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ SNP_ID              : chr [1:983] "abph1.20" "abph1.22" "ae1.3" "ae1.4" ...
##  $ cdv_marker_id       : num [1:983] 5976 5978 6605 6606 6607 ...
##  $ Chromosome          : chr [1:983] "2" "2" "5" "5" ...
##  $ Position            : chr [1:983] "27403404" "27403892" "167889790" "167889682" ...
##  $ alt_pos             : chr [1:983] NA NA NA NA ...
##  $ mult_positions      : chr [1:983] NA NA NA NA ...
##  $ amplicon            : chr [1:983] "abph1" "abph1" "ae1" "ae1" ...
##  $ cdv_map_feature.name: chr [1:983] "AB042260" "AB042260" "ae1" "ae1" ...
##  $ gene                : chr [1:983] "abph1" "abph1" "ae1" "ae1" ...
##  $ candidate/random    : chr [1:983] "candidate" "candidate" "candidate" "candidate" ...
##  $ Genaissance_daa_id  : num [1:983] 8393 8394 8395 8396 8397 ...
##  $ Sequenom_daa_id     : num [1:983] 10474 10475 10477 10478 10479 ...
##  $ count_amplicons     : num [1:983] 1 0 1 0 0 1 1 0 1 0 ...
##  $ count_cmf           : num [1:983] 1 0 1 0 0 1 0 0 1 0 ...
##  $ count_gene          : num [1:983] 1 0 1 0 0 1 1 0 1 0 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   SNP_ID = col_character(),
##   ..   cdv_marker_id = col_double(),
##   ..   Chromosome = col_character(),
##   ..   Position = col_character(),
##   ..   alt_pos = col_character(),
##   ..   mult_positions = col_character(),
##   ..   amplicon = col_character(),
##   ..   cdv_map_feature.name = col_character(),
```

```
##   ..   gene = col_character(),
##   ..   `candidate/random` = col_character(),
##   ..   Genaissance_daa_id = col_double(),
##   ..   Sequenom_daa_id = col_double(),
##   ..   count_amplicons = col_double(),
##   ..   count_cmf = col_double(),
##   ..   count_gene = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
dim(snp_pos_data)
```

```
## [1] 983  15
```

```r
head(snp_pos_data)
```

```
## # A tibble: 6 x 15
##   SNP_ID   cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>            <dbl> <chr>      <chr>     <chr>   <chr>          <chr>
## 1 abph1.20          5976 2          27403404  <NA>    <NA>           abph1
## 2 abph1.22          5978 2          27403892  <NA>    <NA>           abph1
## 3 ae1.3             6605 5          167889790 <NA>    <NA>           ae1
## 4 ae1.4             6606 5          167889682 <NA>    <NA>           ae1
## 5 ae1.5             6607 5          167889821 <NA>    <NA>           ae1
## 6 an1.4             5982 1          240498509 <NA>    <NA>           an1
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>,
## #   count_amplicons <dbl>, count_cmf <dbl>, count_gene <dbl>
```

```r
tail(snp_pos_data)
```

```
## # A tibble: 6 x 15
##   SNP_ID cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>          <dbl> <chr>      <chr>     <chr>   <chr>          <chr>
## 1 zap1.2          3514 2          233128584 <NA>    <NA>           zap1
## 2 zen1.1          3519 unknown    unknown   <NA>    <NA>           zen1
## 3 zen1.2          3520 unknown    unknown   <NA>    <NA>           zen1
## 4 zen1.4          3521 unknown    unknown   <NA>    <NA>           zen1
## 5 zfl2.6          6463 2          12543294  <NA>    <NA>           zfl2
## 6 zmm3.4          3527 9          16966348  <NA>    <NA>           zmm3
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>,
## #   count_amplicons <dbl>, count_cmf <dbl>, count_gene <dbl>
```

```r
# data processing

## inspecting the genotype data group i will be using to subset the data into maize and teosinte

### i used `unique` to Identify available groups to ensure correct filtering
unique(genoty_data$Group)
```

```
##  [1] "TRIPS" "ZDIPL" "ZPERR" "ZLUXR" "ZMHUE" "ZMPBA" "ZMPJA" "ZMXCH" "ZMXCP"
## [10] "ZMXNO" "ZMXNT" "ZMPIL" "ZMXIL" "ZMMLR" "ZMMMR" "ZMMIL"
```

```r
## selecting maize data
maize_genotypes <- genoty_data %>% filter(Group == c("ZMMIL", "ZMMLR", "ZMMMR" ))
```

```
## Warning: There was 1 warning in `filter()`.
## i In argument: `Group == c("ZMMIL", "ZMMLR", "ZMMMR")`.
## Caused by warning in `Group == c("ZMMIL", "ZMMLR", "ZMMMR")`:
## ! longer object length is not a multiple of shorter object length
```

```r
head(maize_genotypes)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 ZDP_0793a Zmm-LR-~ ZMMLR C/G      A/A      T/T   G/G   C/T   C/G   A/G   G/T
## 2 ZDP_0581a Zmm-LR-~ ZMMLR C/C      A/A      T/T   G/G   C/T   C/C   G/G   G/G
## 3 ZDP_0042a Zmm-LR-~ ZMMLR C/C      A/A      T/T   G/G   C/T   C/C   G/G   G/G
## 4 ZDP_0087a Zmm-LR-~ ZMMLR ?/?      A/A      T/T   G/G   T/T   C/C   ?/?   G/G
## 5 ZDP_0168a Zmm-LR-~ ZMMLR C/C      A/A      G/T   A/G   C/T   C/C   G/G   G/G
## 6 ZDP_1098a Zmm-LR-~ ZMMLR G/G      A/A      T/T   G/G   T/T   C/C   G/G   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
dim(maize_genotypes)
```

```
## [1] 525 986
```

```r
unique(maize_genotypes$Group)
```

```
## [1] "ZMMLR" "ZMMMR" "ZMMIL"
```

```r
## selecting teosinte data
teosinte_genotypes <- genoty_data %>% filter(Group == c("ZMPBA", "ZMPIL", "ZMPJA"))
```

```
## Warning: There was 1 warning in `filter()`.
## i In argument: `Group == c("ZMPBA", "ZMPIL", "ZMPJA")`.
## Caused by warning in `Group == c("ZMPBA", "ZMPIL", "ZMPJA")`:
## ! longer object length is not a multiple of shorter object length
```

```r
head(teosinte_genotypes)
```

```
## # A tibble: 6 x 986
##   Sample_ID   JG_OTU Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>       <chr>  <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 S1087       Zmp-B~ ZMPBA G/G      A/A      T/T   G/G   T/T   C/C   A/A   G/G
## 2 S1698       Zmp-B~ ZMPBA C/G      A/A      T/T   A/G   T/T   C/C   A/A   G/G
## 3 TA057       Zmp-B~ ZMPBA G/G      A/A      T/T   G/G   C/T   C/C   A/A   G/T
## 4 TA205       Zmp-B~ ZMPBA C/G      A/A      T/T   G/G   T/T   C/C   A/A   G/T
## 5 TA285       Zmp-B~ ZMPBA G/G      C/C      T/T   G/G   T/T   C/C   A/A   G/T
## 6 TAMex0153.1 Zmp-B~ ZMPBA C/C      A/A      T/T   G/G   C/T   C/C   A/A   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
dim(teosinte_genotypes)
```

```
## [1] 325 986
```

```r
unique(teosinte_genotypes$Group)
```

```
## [1] "ZMPBA" "ZMPJA" "ZMPIL"
```

```r
# transposing genotype files
## I transposed the data so that SNP_IDs become rows, allowing to join with the position map.

### maize genotypes
maize_genotypes_transposed <- t(maize_genotypes)
dim(maize_genotypes_transposed)
```

```
## [1] 986 525
```

```r
dim(maize_genotypes)
```

```
## [1] 525 986
```

```r
maize_genotypes_transposed <- as.data.frame(maize_genotypes_transposed)
colnames(maize_genotypes_transposed) <- maize_genotypes_transposed[1, ] # Set header to Sample IDs
maize_genotypes_transposed <- maize_genotypes_transposed[-1, ] # Remove the extra header row
maize_genotypes_transposed$SNP_ID <- rownames(maize_genotypes_transposed)
rownames(maize_genotypes_transposed) <- NULL
head(maize_genotypes_transposed)
```

```
##              ZDP_0793a            ZDP_0581a            ZDP_0042a
## 1 Zmm-LR-ACOM-usa-NM-2 Zmm-LR-ALAZ-Per-PICS Zmm-LR-ALTI-Arg-HUMA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
```

```
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0087a            ZDP_0168a              ZDP_1098a
## 1 Zmm-LR-ALTI-Bol-TACA Zmm-LR-AMAG-Col-COCO Zmm-LR-AMRH-Per-HUAN-2
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  ?/?                  C/C                    G/G
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  G/T                    T/T
## 6                  G/G                  A/G                    G/G
##              ZDP_0009a            ZDP_0018a            ZDP_1053a
## 1 Zmm-LR-ANCA-Per-CARH Zmm-LR-ANCA-Per-SIHU Zmm-LR-ANCH-Mex-SAND
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  ?/?                  C/G                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##                ZDP_0164a            ZDP_0849a            ZDP_0179a
## 1 Zmm-LR-ANDA-Col-BELE-2 Zmm-LR-APCH-usa-AZ-2 Zmm-LR-APCT-Mex-BASA
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/G                  C/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0181a            ZDP_0856a              ZDP_1476a
## 1 Zmm-LR-APCT-Mex-ESTA Zmm-LR-ARAG-Ven-ANZO-2 Zmm-LR-ARAG-Ven-ANZO-5
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/G                  C/C                    C/G
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  G/T                    T/T
## 6                  G/G                  A/G                    G/G
##                ZDP_0206a            ZDP_0054a         ZDP_0791a
## 1 Zmm-LR-ARAU-Chi-CAUT-2 Zmm-LR-AREQ-Per-COCA Zmm-LR-ARGP-Arg-2
## 2                  ZMMLR                ZMMLR             ZMMLR
## 3                    C/C                  C/C               ?/?
## 4                    A/A                  A/A               ?/?
## 5                    T/T                  T/T               G/G
## 6                    G/G                  G/G               A/A
##              ZDP_0090a            ZDP_0061a            ZDP_0570a
## 1 Zmm-LR-ARGT-Bol-SANT Zmm-LR-ARIZ-Per-BADO Zmm-LR-ARIZ-Per-VIRU
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/G                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##                ZDP_1104a            ZDP_0634a            ZDP_0733b
## 1 Zmm-LR-ARRO-Mex-PLAN-1 Zmm-LR-ARRO-Mex-ZARA Zmm-LR-ASSI-usa-ND-3
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  ?/?                  T/T
## 6                    G/G                  G/G                  G/G
##           ZDP_1012a         ZDP_0154a              ZDP_0994a
## 1 Zmm-LR-AVPI-Bra-2 Zmm-LR-AVTP-Bra-2 Zmm-LR-AYSU-Bol-CHUQ-2
## 2              ZMMLR             ZMMLR                  ZMMLR
```

```
## 3               G/G               C/C                     C/C
## 4               A/A               A/A                     A/A
## 5               T/T               G/T                     T/T
## 6               G/G               A/G                     G/G
##                ZDP_0123b            ZDP_0174a               ZDP_1481a
## 1 Zmm-LR-AYSU-Bol-POTO-1 Zmm-LR-AZUL-Mex-GOME Zmm-LR-AZUL-Mex-MATA-2
## 2                  ZMMLR                ZMMLR                   ZMMLR
## 3                    C/C                  C/C                     C/G
## 4                    A/A                  A/A                     A/A
## 5                    T/T                  T/T                     T/T
## 6                    G/G                  G/G                     G/G
##              ZDP_0368b         BKN_004              ZDP_0563a
## 1 Zmm-LR-BBLA-Ecu-IMBA Zmm-LR-BEAP-2_f Zmm-LR-BHRD-Ecu-LOJA-2
## 2                ZMMLR           ZMMLR                   ZMMLR
## 3                  C/C             C/C                     C/C
## 4                  A/A             A/A                     A/A
## 5                  T/T             T/T                     T/T
## 6                  G/G             G/G                     G/G
##              ZDP_0756b            ZDP_1100a               ZDP_1008b
## 1 Zmm-LR-BLFT-usa-ND-2 Zmm-LR-BOFO-Mex-HUAZ Zmm-LR-BOFO-Mex-SJDP-2
## 2                ZMMLR                ZMMLR                   ZMMLR
## 3                  C/C                  G/G                     C/G
## 4                  A/A                  A/A                     A/A
## 5                  T/T                  T/T                     G/T
## 6                  G/G                  G/G                     A/G
##              ZDP_0639a                  S1823          ZDP_0674b
## 1 Zmm-LR-BOLI-Mex-AGUA Zmm-LR-BOLI-Mex-NOCH-1 Zmm-LR-BRZB-usa-WI
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  ?/?
## 6                  G/G                  G/G                  G/G
##              ZDP_0919a              ZDP_0979b       ZDP_1051b
## 1 Zmm-LR-BSON-Mex-BACA Zmm-LR-CABU-Col-PAMP-2 Zmm-LR-CACA-Mex
## 2                ZMMLR                ZMMLR           ZMMLR
## 3                  C/G                  C/C             C/C
## 4                  A/A                  A/A             A/A
## 5                  T/T                  G/T             T/T
## 6                  G/G                  A/G             G/G
##              ZDP_0350a            ZDP_0954b               ZDP_1478b
## 1 Zmm-LR-CACA-Mex-TOLU Zmm-LR-CACO-Col-SGIL Zmm-LR-CACO-Ven-MERI-2
## 2                ZMMLR                ZMMLR                   ZMMLR
## 3                  C/C                  C/C                     C/C
## 4                  A/A                  A/A                     A/A
## 5                  T/T                  T/T                     T/T
## 6                  G/G                  G/G                     G/G
##           ZDP_0901b            ZDP_0397a              ZDP_0146a
## 1 Zmm-LR-CAIN-Bra-3 Zmm-LR-CAMA-Ecu-MANA Zmm-LR-CAMB-Bol-LAPA-2
## 2             ZMMLR                ZMMLR                   ZMMLR
## 3               C/C                  G/G                     C/C
## 4               A/A                  A/A                     A/A
## 5               T/T                  T/T                     T/T
## 6               G/G                  G/G                     G/G
##              ZDP_0286b              ZDP_0438a                  S1800
```

```
## 1 Zmm-LR-CAND-Ecu-CHON Zmm-LR-CAND-Ecu-MANA-3 Zmm-LR-CANI-Ven-ANZO-1
## 2                ZMMLR                  ZMMLR                  ZMMLR
## 3                  C/G                    C/G                    C/C
## 4                  A/A                    A/A                    A/A
## 5                  T/T                    G/G                    T/T
## 6                  G/G                    A/A                    G/G
##               ZDP_1475a              ZDP_0209a              ZDP_0208a
## 1 Zmm-LR-CANI-Ven-ANZO-4 Zmm-LR-CAPC-Chi-ANTO-1 Zmm-LR-CAPG-Chi-ANTO-2
## 2                  ZMMLR                  ZMMLR                  ZMMLR
## 3                    C/G                    C/C                    C/C
## 4                    A/A                    A/A                    A/A
## 5                    T/T                    T/T                    T/T
## 6                    ?/?                    G/G                    G/G
##               ZDP_0029a            ZDP_0557a            ZDP_0296a
## 1 Zmm-LR-CAPI-Col-RION-2 Zmm-LR-CAQT-Col-ALGE Zmm-LR-CARI-Col-ARAC
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/G                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##               ZDP_1449b            ZDP_0925b            ZDP_0862a
## 1 Zmm-LR-CARI-Col-MONT-2 Zmm-LR-CARI-Ven-ANZO Zmm-LR-CARI-Ven-ESPI
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/G                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    G/T                  T/T                  G/T
## 6                    A/G                  G/G                  A/G
##               ZDP_0047a            ZDP_0451a            ZDP_1042a
## 1 Zmm-LR-CBLA-Arg-HUMA-2 Zmm-LR-CCAT-Ecu-NAPO Zmm-LR-CDEN-Bol-SIGN
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0445a            ZDP_0423a              ZDP_1491a
## 1 Zmm-LR-CELA-Mex-CORT Zmm-LR-CELA-Mex-PROG Zmm-LR-CFLI-Cub-HAVA-2
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/G                  C/C                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  T/T                    G/T
## 6                  G/G                  G/G                    A/G
##              ZDP_0060a            ZDP_0558a            ZDP_0156a
## 1 Zmm-LR-CFMO-Per-HUAM Zmm-LR-CFPT-Per-ACOB Zmm-LR-CFPT-Per-JOSE
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  ?/?                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##           ZDP_1109a              ZDP_1078a            ZDP_0998a
## 1 Zmm-LR-CFPU-Per-1 Zmm-LR-CFPU-Per-TORI-2 Zmm-LR-CHAD-Cub-HOLG
## 2             ZMMLR                  ZMMLR                ZMMLR
## 3               C/C                    C/C                  C/C
## 4               A/A                    A/A                  A/A
## 5               T/T                    T/T                  T/T
```

```
## 6               G/G                     G/G                  G/G
##              ZDP_0513b            ZDP_0854a              ZDP_0125a
## 1 Zmm-LR-CHAD-Hai-PETI Zmm-LR-CHAD-Ven-ARAG Zmm-LR-CHAK-Bol-CHUQ-2
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  C/C                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  G/T                    T/T
## 6                  G/G                  A/G                    G/G
##                ZDP_0089a            ZDP_0632a              ZDP_0646a
## 1 Zmm-LR-CHAK-Bol-COCH-3 Zmm-LR-CHAL-Mex-CHOL Zmm-LR-CHAL-Mex-TEPO
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/G                  C/C                    ?/?
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  T/T                    T/T
## 6                    G/G                  G/G                    G/G
##              ZDP_0981b            ZDP_0969b            ZDP_0214a
## 1 Zmm-LR-CHAP-Mex-ABAJ Zmm-LR-CHAP-Mex-CULI Zmm-LR-CHCL-Chi-ACON
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  ?/?                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0641a            ZDP_0254a              ZDP_0400a
## 1 Zmm-LR-CHCO-Col-LLOR Zmm-LR-CHCO-Col-TADO Zmm-LR-CHCO-Ecu-ESME-2
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  C/G                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  T/T                    G/T
## 6                  G/G                  G/G                    A/G
##                ZDP_1466a            ZDP_0662b              ZDP_0653b
## 1 Zmm-LR-CHCO-Ecu-ESME-5 Zmm-LR-CHER-usa-OK-1 Zmm-LR-CHEY-usa-OK-1
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  G/G                    C/G
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  T/T                    T/T
## 6                    G/G                  G/G                    G/G
##         ZDP_0354a              ZDP_0311a         ZDP_0864b            ZDP_0096a
## 1 Zmm-LR-CHIL-Ecu Zmm-LR-CHIL-Ecu-MERC-2 Zmm-LR-CHIR-Ven-1 Zmm-LR-CHOL-Bol-ELPA
## 2           ZMMLR                  ZMMLR             ZMMLR                ZMMLR
## 3             C/C                    ?/?               C/C                  C/C
## 4             A/A                    A/A               A/A                  A/A
## 5             T/T                    T/T               T/T                  T/T
## 6             G/G                    G/G               G/G                  G/G
##                ZDP_0324b                ZDP_0347b                  S1804
## 1 Zmm-LR-CHOL-Ecu-ZAMO-1 Zmm-LR-CHPI-Per-HUAC-1_s Zmm-LR-CHUL-Per-ACOR-1
## 2                  ZMMLR                    ZMMLR                  ZMMLR
## 3                    G/G                      C/G                    C/C
## 4                    A/A                      A/A                    A/A
## 5                    T/T                      T/T                    T/T
## 6                    G/G                      G/G                    G/G
##                ZDP_1206b            ZDP_0472a              ZDP_0278a
## 1 Zmm-LR-CHUL-Per-HUAN-1 Zmm-LR-CHUN-Per-CHAG Zmm-LR-CHUN-Per-QUIL
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    C/G
```

```
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  G/T
## 6                    G/G                  G/G                  A/G
##                ZDP_0220a            ZDP_0355a              ZDP_1002b
## 1 Zmm-LR-CHUP-Chi-TARA-2 Zmm-LR-CHUP-Ecu-PUEM Zmm-LR-CHUS-Bol-COCH-2
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  T/T                    T/T
## 6                    G/G                  G/G                    G/G
##                ZDP_0421a            ZDP_0352a            ZDP_0342a
## 1 Zmm-LR-CLAT-Ecu-TUNG-2 Zmm-LR-CLAV-Col-PAST Zmm-LR-CNGG-Ecu-CUSU
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  ?/?                  T/T
## 6                    G/G                  G/G                  G/G
##           ZDP_0992a              ZDP_0792a            ZDP_0682b
## 1 Zmm-LR-COCH-Arg-2 Zmm-LR-COCP-usa-NM-2_s Zmm-LR-COLO-usa-CO-3
## 2             ZMMLR                  ZMMLR                ZMMLR
## 3               C/C                    G/G                  G/G
## 4               A/A                    A/A                  A/A
## 5               T/T                    T/T                  G/T
## 6               G/G                    G/G                  A/G
##              ZDP_0489a            ZDP_0248a            ZDP_1023a
## 1 Zmm-LR-COMI-Gua-MIXC Zmm-LR-COMI-Mex-FINC Zmm-LR-COMI-Mex-RITA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  G/G                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  G/T                  G/T                  T/T
## 6                  A/G                  A/G                  G/G
##              ZDP_0832a            ZDP_0908a            ZDP_0365b
## 1 Zmm-LR-COMU-Col-VERS Zmm-LR-COMU-Ven-LARA-2 Zmm-LR-COND-Ecu-GUAL
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##                ZDP_0412a            ZDP_0402a            ZDP_0649a
## 1 Zmm-LR-CONE-Mex-JERO-2 Zmm-LR-CONE-Mex-ZACU Zmm-LR-CONI-Mex-IXTL
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/G                  C/C                  G/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  ?/?                  G/G
##              ZDP_0308a            ZDP_0434a            ZDP_1170a
## 1 Zmm-LR-CONI-Mex-TLAC Zmm-LR-CONN-Mex-JARA Zmm-LR-CONN-Mex-JUAN-2
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  G/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  ?/?                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0002a            ZDP_0136a            ZDP_0141a
## 1 Zmm-LR-CONN-Mex-TEPE Zmm-LR-CORA-Bol-TALA Zmm-LR-CORB-Bol-PAND
```

```
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  G/G                  G/G
## 6                  G/G                  A/A                  A/A
##              ZDP_0139a            ZDP_0140a            ZDP_0135a
## 1 Zmm-LR-CORO-Bol-BENI Zmm-LR-CORO-Bol-PAND Zmm-LR-CORO-Bol-TALA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  G/G                  G/G                  T/T
## 6                  A/A                  A/A                  G/G
##                ZDP_0938a            ZDP_0880b            ZDP_0341a
## 1 Zmm-LR-COSC-Mex-CUCH-1 Zmm-LR-COSC-Mex-ELOT Zmm-LR-COST-Col-MONT
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/G                  C/C                  C/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0898b            ZDP_0936a            ZDP_0177a
## 1 Zmm-LR-COST-Ven-APUR Zmm-LR-COST-Ven-YARA-1 Zmm-LR-CRCH-Mex-GOME
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  G/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0171a            ZDP_1495b       ZDP_0211a
## 1 Zmm-LR-CRCH-Mex-MADE Zmm-LR-CRNO-Chi-CAUT-3 Zmm-LR-CRSC-Chi
## 2                ZMMLR                ZMMLR           ZMMLR
## 3                  C/G                  C/C             C/C
## 4                  A/A                  A/A             A/A
## 5                  T/T                  T/T             ?/?
## 6                  G/G                  G/G             G/G
##              ZDP_0205a       ZDP_0204a            ZDP_1028a         ZDP_0036a
## 1 Zmm-LR-CRSC-Chi-OSOR Zmm-LR-CRSG-Chi Zmm-LR-CRSG-Chi-NUBL Zmm-LR-CRSS-Arg-1
## 2                ZMMLR           ZMMLR                ZMMLR             ZMMLR
## 3                  C/C             C/C                  C/C               C/C
## 4                  A/A             A/A                  A/A               A/A
## 5                  G/T             T/T                  T/T               T/T
## 6                  A/G             G/G                  G/G               G/G
##           ZDP_0063a         ZDP_1060a             S1831         ZDP_1497b
## 1 Zmm-LR-CRST-Bra-1 Zmm-LR-CRST-Bra-4 Zmm-LR-CRVR-Bra-1 Zmm-LR-CRVR-Bra-4
## 2             ZMMLR             ZMMLR             ZMMLR             ZMMLR
## 3               G/G               C/C               C/C               C/C
## 4               A/A               A/A               A/A               A/A
## 5               T/T               T/T               T/T               T/T
## 6               G/G               G/G               G/G               G/G
##           ZDP_0922a       ZDP_1041b       ZDP_0389a            ZDP_1215b
## 1 Zmm-LR-CTSE-Uru-1 Zmm-LR-CTSP-Arg Zmm-LR-CTTN-Frg Zmm-LR-CTTN-Sur-LELY
## 2             ZMMLR           ZMMLR           ZMMLR                ZMMLR
## 3               C/C             C/G             G/G                  C/C
## 4               A/A             A/A             A/A                  A/A
## 5               T/T             T/T             T/T                  T/T
## 6               G/G             G/G             G/G                  G/G
```

```
##                ZDP_1213a            ZDP_0115b         ZDP_0351a
## 1 Zmm-LR-CTTO-Bol-CANA-3 Zmm-LR-CTTO-Bol-SANT Zmm-LR-CTTO-Bra-3
## 2                  ZMMLR                ZMMLR             ZMMLR
## 3                    C/G                  C/C               ?/?
## 4                    A/A                  A/A               A/A
## 5                    T/T                  T/T               G/T
## 6                    G/G                  G/G               A/G
##           ZDP_0037a         ZDP_0942b              ZDP_0313b
## 1 Zmm-LR-CTTS-Arg-1 Zmm-LR-CTTS-Uru-2 Zmm-LR-CTUS-Ecu-MANA-1
## 2             ZMMLR             ZMMLR                  ZMMLR
## 3               C/C               C/C                    C/G
## 4               A/A               A/A                    A/A
## 5               T/T               G/T                    G/T
## 6               G/G               A/G                    A/G
##              ZDP_0041a              ZDP_0193a            ZDP_1075a
## 1 Zmm-LR-CULL-Arg-HUMA Zmm-LR-CURA-Chi-COLI-2 Zmm-LR-CURG-Chi-HUEL
## 2                ZMMLR                  ZMMLR                ZMMLR
## 3                  C/C                    C/C                  C/C
## 4                  A/A                    A/A                  A/A
## 5                  T/T                    ?/?                  T/T
## 6                  G/G                    G/G                  G/G
##              ZDP_0271a            ZDP_0273a            ZDP_0288a
## 1 Zmm-LR-CUZA-Per-CALC Zmm-LR-CUZA-Per-SSEB Zmm-LR-CUZC-Ecu-BIBL
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0549a            ZDP_0580a            ZDP_0155a
## 1 Zmm-LR-CUZC-Ecu-POMA Zmm-LR-CUZC-Per-HUAM Zmm-LR-CUZG-Per-CAJA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0923a            ZDP_0943b         ZDP_1121a         ZDP_1179a
## 1 Zmm-LR-CYFL-Ven-ANZO Zmm-LR-CYFL-Ven-COJE Zmm-LR-DBRG-Bra-2 Zmm-LR-DBRG-Bra-5
## 2                ZMMLR                ZMMLR             ZMMLR             ZMMLR
## 3                  C/C                  C/C               C/G               G/G
## 4                  A/A                  A/A               A/A               A/A
## 5                  T/T                  G/T               T/T               T/T
## 6                  G/G                  A/G               G/G               G/G
##           ZDP_1013a            ZDP_0198a         ZDP_0910a         ZDP_0952a
## 1 Zmm-LR-DENC-Chi-1 Zmm-LR-DENC-Chi-MALL Zmm-LR-DENP-Bra-3 Zmm-LR-DRGL-Bra-1
## 2             ZMMLR                ZMMLR             ZMMLR             ZMMLR
## 3               C/C                  C/C               C/G               G/G
## 4               A/A                  A/A               A/A               A/A
## 5               T/T                  T/T               G/T               G/T
## 6               G/G                  ?/?               A/G               A/G
##           ZDP_1176a         ZDP_1116b       ZDP_0874b              ZDP_0955b
## 1 Zmm-LR-DRGL-Bra-4 Zmm-LR-DRGR-Bra-3 Zmm-LR-DUDN-Mex Zmm-LR-DUDN-Mex-NORI-1
## 2             ZMMLR             ZMMLR           ZMMLR                  ZMMLR
## 3               G/G               ?/?             C/C                    C/C
## 4               A/A               A/A             A/A                    A/A
```

```
## 5               T/T               T/T             G/G                     T/T
## 6               G/G               G/G             A/A                     G/G
##                ZDP_0203a            ZDP_1097a            ZDP_1081a
## 1 Zmm-LR-DUGB-Chi-MALL-2 Zmm-LR-DULJ-Mex-MEXQ Zmm-LR-DULJ-Mex-YAHU
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/G                  C/C                  C/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##                ZDP_1465a                  S1835                  S1553
## 1 Zmm-LR-DZIT-Gua-ASUN-2 Zmm-LR-DZIT-Gua-TENC-1 Zmm-LR-DZIT-Mex-MUJE-1
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    G/T                  T/T                    G/T
## 6                    A/G                  G/G                    A/G
##                ZDP_0307a              ZDP_0371a            ZDP_1093b
## 1 Zmm-LR-ECAR-Mar-BASS-1 Zmm-LR-ECAR-Mar-PIER-2 Zmm-LR-ELOC-Mex-SMHI
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0824a            ZDP_0610a            ZDP_1076a
## 1 Zmm-LR-ELOO-Mex-JALP Zmm-LR-ELOO-Mex-TEPA Zmm-LR-ELOO-Mex-ZAPO
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/G                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  G/T                  T/T
## 6                  G/G                  A/G                  G/G
##              ZDP_1062b            ZDP_0147a            ZDP_0137a
## 1 Zmm-LR-ELOS-Mex-PENI Zmm-LR-ENAN-Bol-PAND Zmm-LR-ENAN-Bol-TALA-2
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0481a            ZDP_0469a            ZDP_0455a
## 1 Zmm-LR-FRIA-Gua-CAND Zmm-LR-FRIA-Gua-SLOR Zmm-LR-GALL-Ecu-ELOR
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/G                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  ?/?                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0309a          ZDP_0790a            ZDP_0800a
## 1 Zmm-LR-GALL-Ecu-PINA Zmm-LR-GEHU-Can-MB Zmm-LR-GEHU-usa-ND-2
## 2                ZMMLR              ZMMLR                ZMMLR
## 3                  ?/?                C/C                  G/G
## 4                  A/A                A/A                  A/A
## 5                  T/T                T/T                  T/T
## 6                  G/G                G/G                  ?/?
##           ZDP_0802a              ZDP_1493b            ZDP_0656b
## 1 Zmm-LR-GGLO-usa-3 Zmm-LR-GORD-Mex-GUER-2 Zmm-LR-GOUR-usa-TX-1
## 2             ZMMLR                  ZMMLR                ZMMLR
```

```
## 3               C/C                     C/G                  G/G
## 4               A/A                     A/A                  A/A
## 5               T/T                     T/T                  T/T
## 6               G/G                     G/G                  G/G
##              ZDP_0830a            ZDP_0008a            ZDP_0837a
## 1 Zmm-LR-GOUR-usa-TX-4 Zmm-LR-GRAN-Per-MACA Zmm-LR-GUAR-Ven-GUAR
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##               ZDP_0306a            ZDP_0575a            ZDP_1101b
## 1 Zmm-LR-GUIR-Col-VALL-2 Zmm-LR-HAIY-Hai-JOSA Zmm-LR-HAIY-Hai-MORO-3
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    ?/?                  C/G                    C/G
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  T/T                    T/T
## 6                    G/G                  G/G                    G/G
##               ZDP_0216a            ZDP_1216b            ZDP_0676b
## 1 Zmm-LR-HART-Chi-TARA-2 Zmm-LR-HART-Chi-TARA-5 Zmm-LR-HAVA-usa-AZ-2
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    C/G                  G/G
## 4                    A/A                    A/A                  A/A
## 5                    T/T                    T/T                  G/G
## 6                    G/G                    G/G                  A/A
##           ZDP_1120a         ZDP_0739a            ZDP_0867b
## 1 Zmm-LR-HICK-Bra-1 Zmm-LR-HICK-usa-2 Zmm-LR-HOPI-usa-AZ-3
## 2             ZMMLR             ZMMLR                ZMMLR
## 3               ?/?               G/G                  G/G
## 4               A/A               A/A                  A/A
## 5               T/T               T/T                  T/T
## 6               G/G               G/G                  G/G
##               ZDP_0594a            ZDP_0398a          ZDP_0829b
## 1 Zmm-LR-HUAC-Per-HUAC-3 Zmm-LR-HUAD-Ecu-IMBA-1 Zmm-LR-HUAL-usa-AZ
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    C/C                  G/G
## 4                    A/A                    A/A                  A/A
## 5                    G/T                    T/T                  T/T
## 6                    A/G                    G/G                  G/G
##              ZDP_0616a            ZDP_0275b            ZDP_0020a
## 1 Zmm-LR-HUAR-Per-LIRO Zmm-LR-HUAV-Per-CALC Zmm-LR-HUAY-Per-HUAY
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0887b       ZDP_0092b              ZDP_1200a
## 1 Zmm-LR-HUEV-Ven-TACH Zmm-LR-HUIM-Bol Zmm-LR-IMBR-Col-IPIA-1
## 2                ZMMLR           ZMMLR                  ZMMLR
## 3                  C/C             C/C                    C/C
## 4                  A/A             A/A                    A/A
## 5                  T/T             T/T                    T/T
## 6                  G/G             G/G                    G/G
##              ZDP_0519a            ZDP_0751a            ZDP_1092b
```

```
## 1 Zmm-LR-IMBR-Gua-CALZ Zmm-LR-ISLE-usa-NM-2 Zmm-LR-JALA-Mex-JALA-3
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  G/G                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  T/T                    T/T
## 6                  G/G                  G/G                    G/G
##               ZDP_0566a            ZDP_0795a                  S1837
## 1 Zmm-LR-JALA-Mex-TOCO-2 Zmm-LR-JEME-usa-NM-2 Zmm-LR-KARA-Bol-CHUQ-1
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    ?/?                  C/G                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    ?/?                  ?/?                    T/T
## 6                    G/G                  A/A                    G/G
##               ZDP_0129b            ZDP_0071a              ZDP_0375a
## 1 Zmm-LR-KARA-Bol-CHUQ-4 Zmm-LR-KCEL-Bol-COCH Zmm-LR-KCEL-Ecu-LOJA-1
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  G/T                    T/T
## 6                    G/G                  A/G                    G/G
##              ZDP_0576a            ZDP_1205b         ZDP_0754b
## 1 Zmm-LR-KCUL-Per-ACOR Zmm-LR-KCUL-Per-JERO-3 Zmm-LR-KRUG-usa-1
## 2                ZMMLR                ZMMLR             ZMMLR
## 3                  C/C                  C/C               C/G
## 4                  A/A                  A/A               A/A
## 5                  T/T                  T/T               T/T
## 6                  G/G                  G/G               G/G
##           ZDP_0840a            ZDP_1043b            ZDP_0742a       ZDP_0966b
## 1 Zmm-LR-KRUG-usa-4 Zmm-LR-KULL-Bol-POTO-2 Zmm-LR-LAGU-usa-NM-2 Zmm-LR-LENH-Bra
## 2             ZMMLR                ZMMLR                ZMMLR           ZMMLR
## 3               C/C                  C/C                  G/G             C/G
## 4               A/A                  A/A                  A/A             A/A
## 5               T/T                  T/T                  G/T             T/T
## 6               G/G                  G/G                  A/G             ?/?
##              ZDP_0770a            ZDP_0495a              ZDP_0562a
## 1 Zmm-LR-LONG-Can-ON-2 Zmm-LR-MAIZ-Col-PAST Zmm-LR-MARA-Per-HUAN-2
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  C/G                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  T/T                    T/T
## 6                  G/G                  G/G                    G/G
##               ZDP_0221b            ZDP_0995a            ZDP_0294a
## 1 Zmm-LR-MARC-Chi-TARA-1 Zmm-LR-MCAJ-Per-CAJA Zmm-LR-MCAN-Per-LIMA
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  ?/?
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  ?/?
## 6                    G/G                  G/G                  A/G
##              ZDP_0537b            ZDP_0320a            ZDP_0763a
## 1 Zmm-LR-MCHL-Per-OLLE Zmm-LR-MCHN-Ecu-SANG-1 Zmm-LR-MESP-usa-NM-1
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  ?/?                  T/T
```

```
## 6                 G/G                   G/G                 ?/?
##            ZDP_0677b          ZDP_0759b            ZDP_0322a
## 1 Zmm-LR-MEXJ-usa-VA Zmm-LR-MINN-usa-MN Zmm-LR-MISH-Ecu-URCU
## 2              ZMMLR              ZMMLR                ZMMLR
## 3                C/G                C/C                  C/C
## 4                A/A                A/A                  A/A
## 5                T/T                T/T                  T/T
## 6                G/G                G/G                  G/G
##              ZDP_0630a         ZDP_0868b              ZDP_0548a
## 1 Zmm-LR-MIXT-Mex-SJDG Zmm-LR-MIYD-usa-2 Zmm-LR-MNTY-Mex-TAPA-1
## 2                ZMMLR             ZMMLR                  ZMMLR
## 3                  C/G               C/C                    C/C
## 4                  A/A               A/A                    A/A
## 5                  T/T               T/T                    T/T
## 6                  G/G               G/G                    G/G
##                ZDP_0611a            ZDP_0592a            ZDP_0743b
## 1 Zmm-LR-MOCH-Per-LAMB-1 Zmm-LR-MOCH-Per-PICS Zmm-LR-MOHA-usa-AZ-1
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  G/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  G/T                  G/T
## 6                    G/G                  A/G                  A/G
##                    S1810            ZDP_0032a              ZDP_0346a
## 1 Zmm-LR-MONT-Col-PUER-1 Zmm-LR-MONT-Col-UNIO Zmm-LR-MONT-Ecu-IMBA-2
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    G/T                  T/T                    T/T
## 6                    A/G                  G/G                    G/G
##                ZDP_1212a             ZDP_0103b            ZDP_0004a
## 1 Zmm-LR-MORA-Bol-IRUP-2 Zmm-LR-MORA-Bol-IVO-2 Zmm-LR-MORO-Per-LUYA
## 2                  ZMMLR                 ZMMLR                ZMMLR
## 3                    C/G                   C/G                  C/C
## 4                    A/A                   A/A                  A/A
## 5                    G/T                   T/T                  T/T
## 6                    A/G                   G/G                  G/G
##              ZDP_0582a              ZDP_0990b       ZDP_0066a         ZDP_0466a
## 1 Zmm-LR-MORO-Per-OTUZ Zmm-LR-MOTO-Mex-MOTO-3 Zmm-LR-MRTI-Bol Zmm-LR-MRTI-Bra-3
## 2                ZMMLR                 ZMMLR           ZMMLR             ZMMLR
## 3                  C/C                   C/G             C/C               ?/?
## 4                  A/A                   A/A             A/A               A/A
## 5                  T/T                   T/T             G/T               T/T
## 6                  G/G                   G/G             A/G               G/G
##         ZDP_0065a            ZDP_0643a              ZDP_1471a
## 1 Zmm-LR-MRTP-Bol Zmm-LR-MUSH-Mex-MACA Zmm-LR-MUSH-Mex-TANA-3
## 2           ZMMLR                ZMMLR                  ZMMLR
## 3             C/C                  C/C                    C/G
## 4             A/A                  A/A                    A/A
## 5             T/T                  ?/?                    T/T
## 6             G/G                  G/G                    G/G
##              ZDP_0774b            ZDP_0449a            ZDP_0585a
## 1 Zmm-LR-NAVA-usa-AZ-3 Zmm-LR-NDCA-Gua-CUES Zmm-LR-NDCA-Gua-RECU
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  G/G                  C/G                  C/C
```

```
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0482a            ZDP_0251a         ZDP_1050b
## 1 Zmm-LR-NDCH-Gua-CANO Zmm-LR-NECH-Mex-NAVE Zmm-LR-NEFL-usa-2
## 2                ZMMLR                ZMMLR             ZMMLR
## 3                  G/G                  C/G               C/C
## 4                  A/A                  A/A               A/A
## 5                  T/T                  G/T               T/T
## 6                  G/G                  G/G               G/G
##              ZDP_0476a         ZDP_0926a            ZDP_0383a
## 1 Zmm-LR-NEGR-Mex-BOCA Zmm-LR-NEGR-Ven-2 Zmm-LR-NTAB-Gua-BUEN
## 2                ZMMLR             ZMMLR                ZMMLR
## 3                  C/C               G/G                  C/C
## 4                  A/A               A/A                  A/A
## 5                  T/T               T/T                  T/T
## 6                  G/G               G/G                  G/G
##              ZDP_0426a              ZDP_0237a            ZDP_0392a
## 1 Zmm-LR-NTAB-Gua-QUEZ Zmm-LR-NTAL-Mex-BEJU-2 Zmm-LR-NTBA-Gua-CHIQ
## 2                ZMMLR                  ZMMLR                ZMMLR
## 3                  C/G                    C/C                  C/G
## 4                  A/A                    A/A                  A/A
## 5                  G/G                    T/T                  T/T
## 6                  A/A                    G/G                  G/G
##                ZDP_0437a            ZDP_0815a                  S1433
## 1 Zmm-LR-NTBB-Gua-PROG-1 Zmm-LR-NTEL-Mex-DZIT Zmm-LR-NTEL-Mex-PUEB-1
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##                ZDP_1171a          ZDP_0847a            ZDP_0050a
## 1 Zmm-LR-NTEL-Mex-TIKO-2 Zmm-LR-NWDE-usa-CO Zmm-LR-OKEE-Arg-HUMA
## 2                  ZMMLR              ZMMLR                ZMMLR
## 3                    C/C                C/C                  C/G
## 4                    A/A                A/A                  A/A
## 5                    T/T                T/T                  T/T
## 6                    G/G                G/G                  G/G
##                ZDP_0232a            ZDP_1014a            ZDP_0446a
## 1 Zmm-LR-OLTL-Mex-TERA-2 Zmm-LR-OLTL-Mex-TOTO Zmm-LR-OLTN-Gua-CONC
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/G                  C/G                  C/G
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  G/T                  T/T
## 6                    G/G                  A/G                  G/G
##              ZDP_0382a            ZDP_0250a                  S1567
## 1 Zmm-LR-OLTN-Gua-SIBI Zmm-LR-OLTN-Mex-CHIL Zmm-LR-ONAV-Mex-GUAD-1
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0939b            ZDP_0506a            ZDP_0578a
## 1 Zmm-LR-ONAV-Mex-TUBU Zmm-LR-PAGA-Per-CHUL Zmm-LR-PAJA-Mex-JUAN-2
```

```
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  C/C                    C/G
## 4                  A/A                  A/A                    A/A
## 5                  ?/?                  G/T                    T/T
## 6                  G/G                  A/G                    G/G
##              ZDP_0850b          ZDP_0781b              ZDP_1000a
## 1 Zmm-LR-PAPA-usa-AZ-2 Zmm-LR-PAPP-usa-AZ Zmm-LR-PARO-Per-JERO-2
## 2                ZMMLR              ZMMLR                  ZMMLR
## 3                  G/G                C/G                    C/C
## 4                  A/A                A/A                    A/A
## 5                  T/T                T/T                    T/T
## 6                  G/G                ?/?                    G/G
##                ZDP_0175b            ZDP_0173a              ZDP_1011a
## 1 Zmm-LR-PATC-Mex-GOME-1 Zmm-LR-PATC-Mex-MADE Zmm-LR-PATG-Bol-LAPA-2
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/G                  C/C                    C/C
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  T/T                    T/T
## 6                    G/G                  G/G                    G/G
##                ZDP_0086b            ZDP_0310a              ZDP_0328a
## 1 Zmm-LR-PATI-Bol-POTO-1 Zmm-LR-PATI-Ecu-MOCH Zmm-LR-PATO-Mex-TOLU-1
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  C/C                    G/G
## 4                    A/A                  A/A                    A/A
## 5                    G/G                  T/T                    ?/?
## 6                    G/G                  G/G                    G/G
##              ZDP_0574a            ZDP_0573a            ZDP_0603a
## 1 Zmm-LR-PEPI-Mex-IGUA Zmm-LR-PEPI-Mex-TLAL-2 Zmm-LR-PERL-Per-HUAU
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/G                  ?/?
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0098a            ZDP_0097a            ZDP_0269b
## 1 Zmm-LR-PERO-Bol-ELPA Zmm-LR-PERO-Bol-VILL Zmm-LR-PIRA-Col-MESI
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  G/T                  T/T
## 6                  G/G                  A/G                  G/G
##           ZDP_0863a            ZDP_1071a            ZDP_0295a
## 1 Zmm-LR-PIRA-Ven-1 Zmm-LR-PIRA-Ven-ATER-2 Zmm-LR-PIRI-Per-YURI-1
## 2             ZMMLR                ZMMLR                ZMMLR
## 3               C/C                  G/G                  C/C
## 4               A/A                  A/A                  A/A
## 5               G/T                  ?/?                  T/T
## 6               A/G                  A/A                  G/G
##              ZDP_0473a            ZDP_0127b            ZDP_0108a
## 1 Zmm-LR-PIRN-Col-BERR Zmm-LR-PISA-Bol-CHUQ Zmm-LR-PISA-Bol-LAPA-2
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/G                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  G/T                  T/T                  T/T
## 6                  A/G                  G/G                  G/G
```

```
##                ZDP_1485a              ZDP_1492b            ZDP_0106b
## 1 Zmm-LR-PISA-Bol-TARI-2 Zmm-LR-PISC-Per-JERO-3 Zmm-LR-POJC-Bol-CUEV
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    C/C                  C/C
## 4                    A/A                    A/A                  A/A
## 5                    T/T                    T/T                  T/T
## 6                    G/G                    G/G                  G/G
##                ZDP_0291a              ZDP_0453a                  BKN_001
## 1 Zmm-LR-POJC-Ecu-NAPO-1 Zmm-LR-POJC-Ecu-NAPO-4 Zmm-LR-POLL-Col-GACH-3_f
## 2                  ZMMLR                  ZMMLR                    ZMMLR
## 3                    C/C                    C/C                      C/C
## 4                    A/A                    A/A                      A/A
## 5                    G/T                    G/G                      T/T
## 6                    A/G                    A/A                      G/G
##              ZDP_0268b              ZDP_1494b            ZDP_0861b
## 1 Zmm-LR-POLL-Col-MANT Zmm-LR-POLL-Col-TIRI-2 Zmm-LR-POLL-Ven-PLAY
## 2                ZMMLR                  ZMMLR                ZMMLR
## 3                  C/C                    C/C                  C/C
## 4                  A/A                    A/A                  A/A
## 5                  T/T                    T/T                  T/T
## 6                  G/G                    G/G                  G/G
##                ZDP_0091a          ZDP_0725b       ZDP_0728a
## 1 Zmm-LR-PORO-Bol-SIGN-2 Zmm-LR-POTA-usa-KS Zmm-LR-PSAL-usa
## 2                  ZMMLR              ZMMLR           ZMMLR
## 3                    C/C                ?/?             G/G
## 4                    A/A                A/A             A/A
## 5                    T/T                T/T             T/T
## 6                    G/G                G/G             G/G
##              ZDP_0698b            ZDP_0749a            ZDP_0348a
## 1 Zmm-LR-PUEB-usa-AZ-3 Zmm-LR-PUHK-usa-AZ-2 Zmm-LR-PUYA-Col-PINT
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  ?/?                  C/G                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  G/T                  T/T
## 6                  G/G                  A/G                  G/G
##              ZDP_0844a            ZDP_0976a              ZDP_0834a
## 1 Zmm-LR-PUYA-Ven-BOMB Zmm-LR-PUYG-Col-OCAN Zmm-LR-PUYG-Ven-BAJO_s
## 2                ZMMLR                ZMMLR                  ZMMLR
## 3                  C/C                  C/C                    C/C
## 4                  A/A                  A/A                    A/A
## 5                  T/T                  T/T                    T/T
## 6                  G/G                  G/G                    G/G
##              ZDP_0809b            ZDP_0542a            ZDP_0532a
## 1 Zmm-LR-PUYG-Ven-MONT Zmm-LR-QUIE-Gua-MALA Zmm-LR-QUIL-Gua-MALA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/G
## 4                  A/A                  A/A                  A/A
## 5                  G/T                  ?/?                  ?/?
## 6                  A/G                  G/G                  G/G
##              ZDP_0941b         ZDP_0744a         ZDP_0812a
## 1 Zmm-LR-RATN-Mex-CONR Zmm-LR-RDYD-usa-2 Zmm-LR-RDYD-usa-5
## 2                ZMMLR              ZMMLR              ZMMLR
## 3                  C/G                C/G                C/G
## 4                  A/A                A/A                A/A
```

```
## 5                  T/T               T/T               T/T
## 6                  G/G               G/G               G/G
##                ZDP_0957b                  S1819            ZDP_0159a
## 1 Zmm-LR-REVE-Mex-ABAJ-1 Zmm-LR-REVE-Mex-TUXP-1 Zmm-LR-RIEN-Per-COND
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    G/G                  C/C
## 4                    A/A                    A/A                  A/A
## 5                    G/T                    T/T                  T/T
## 6                    A/G                    G/G                  G/G
##            ZDP_0748b            ZDP_0301a            ZDP_0022a
## 1 Zmm-LR-RUTH-Can-SK Zmm-LR-RUVA-Ecu-YARU Zmm-LR-RZOR-Per-SIHU
## 2              ZMMLR                ZMMLR                ZMMLR
## 3                ?/?                  C/C                  C/C
## 4                A/A                  A/A                  A/A
## 5                T/T                  T/T                  T/T
## 6                G/G                  G/G                  G/G
##                    S1815            ZDP_0366b            ZDP_0587a
## 1 Zmm-LR-SABA-Col-TOLE-1 Zmm-LR-SABA-Ecu-CALD Zmm-LR-SABA-Ecu-YARU
## 2                  ZMMLR                ZMMLR                ZMMLR
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0873b            ZDP_0806a            ZDP_0490a
## 1 Zmm-LR-SABA-Ven-ESCA Zmm-LR-SABA-Ven-RESG Zmm-LR-SALP-Gua-BARR
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0767a            ZDP_0011a            ZDP_0618a
## 1 Zmm-LR-SAPO-usa-IA-1 Zmm-LR-SARC-Per-RANR-1 Zmm-LR-SCRO-usa-VI
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##           ZDP_0340a          ZDP_0974b            ZDP_0609a
## 1 Zmm-LR-SDPA-Bra-1 Zmm-LR-SDRG-Bra-1 Zmm-LR-SERJ-Mex-ATEM
## 2             ZMMLR              ZMMLR                ZMMLR
## 3               C/C                C/G                  C/C
## 4               A/A                A/A                  A/A
## 5               G/T                T/T                  T/T
## 6               A/G                G/G                  G/G
##                ZDP_0628a              ZDP_0505a            ZDP_0509b
## 1 Zmm-LR-SERJ-Mex-JUAN-3 Zmm-LR-SERM-Mex-YACO-2 Zmm-LR-SERR-Gua-SPED
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    C/C                  C/C
## 4                    A/A                    A/A                  A/A
## 5                    T/T                    T/T                  T/T
## 6                    G/G                    G/G                  G/G
##                ZDP_1467a            ZDP_0545a            ZDP_0014a
## 1 Zmm-LR-SERR-Gua-SSEB-3 Zmm-LR-SGRH-Per-COLC Zmm-LR-SHAJ-Per-ACO-2
## 2                  ZMMLR                ZMMLR                ZMMLR
```

```
## 3                    C/C                  C/C                  C/C
## 4                    A/A                  A/A                  A/A
## 5                    T/T                  T/T                  T/T
## 6                    G/G                  G/G                  G/G
##              ZDP_0704b          ZDP_0688b            ZDP_0540b
## 1 Zmm-LR-SHOE-usa-MO-2 Zmm-LR-SKFL-Can-SK Zmm-LR-SMAR-Gua-COMI
## 2                ZMMLR              ZMMLR                ZMMLR
## 3                  C/C                C/C                  C/G
## 4                  A/A                A/A                  A/A
## 5                  T/T                T/T                  T/T
## 6                  G/G                G/G                  G/G
##              ZDP_0491a          ZDP_0691b          ZDP_0755b
## 1 Zmm-LR-SMAR-Gua-SPED Zmm-LR-SQUA-usa-CO Zmm-LR-SUPG-usa-IA
## 2                ZMMLR              ZMMLR              ZMMLR
## 3                  C/G                C/C                C/C
## 4                  A/A                A/A                A/A
## 5                  T/T                ?/?                T/T
## 6                  G/G                G/G                G/G
##              ZDP_1069a            ZDP_1089b            ZDP_0462a
## 1 Zmm-LR-TABL-Mex-LRIV Zmm-LR-TABL-Mex-TUXP Zmm-LR-TABP-Mex-PALM
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/G                  ?/?
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  G/T
## 6                  G/G                  G/G                  A/G
##              ZDP_1029b            ZDP_0236a            ZDP_0447a
## 1 Zmm-LR-TEHU-Mex-ALVA Zmm-LR-TEHU-Mex-SANA Zmm-LR-TEPE-Gua-ASIN
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  G/T                  G/T
## 6                  G/G                  A/G                  A/G
##              ZDP_0492a            ZDP_0486a            ZDP_0750a
## 1 Zmm-LR-TEPE-Gua-PANZ Zmm-LR-TEPE-Mex-JICA Zmm-LR-TESU-usa-NM-1
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/C                  G/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##              ZDP_0485a                 S1526              ZDP_1472a
## 1 Zmm-LR-TOCH-Mex-AMAT Zmm-LR-TOCH-Mex-COMA-1 Zmm-LR-TOCH-Mex-COMA-4
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/C                  C/G
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  ?/?
## 6                  G/G                  G/G                  G/G
##            ZDP_0695b            ZDP_1464a            ZDP_1039a
## 1 Zmm-LR-TTHB-usa-ND Zmm-LR-TUSI-Ecu-ZAMO-2 Zmm-LR-TUSN-Cub-HOLG-2
## 2              ZMMLR                ZMMLR                ZMMLR
## 3                C/C                  C/G                  G/G
## 4                A/A                  A/A                  A/A
## 5                T/T                  T/T                  T/T
## 6                G/G                  G/G                  G/G
##               ZDP_1099b            ZDP_0807b            ZDP_1190a
```

```
## 1 Zmm-LR-TUSN-Tri-SANG-2 Zmm-LR-TUSN-Ven-CHUR Zmm-LR-TUXN-Mex-APOL-2
## 2                  ZMMLR                ZMMLR                  ZMMLR
## 3                    C/C                  G/G                    G/G
## 4                    A/A                  A/A                    A/A
## 5                    T/T                  G/T                    T/T
## 6                    G/G                  A/G                    G/G
##                    S1439              ZDP_0950b              ZDP_0303a
## 1 Zmm-LR-TUXN-Mex-JANO-1 Zmm-LR-TUXN-Mex-LLER-2 Zmm-LR-TUXP-Ecu-MANA-1
## 2                  ZMMLR                  ZMMLR                  ZMMLR
## 3                    C/C                    C/G                    C/G
## 4                    A/A                    A/A                    A/A
## 5                    T/T                    T/T                    T/T
## 6                    G/G                    G/G                    G/G
##              ZDP_0443a            ZDP_0488a            ZDP_0660b
## 1 Zmm-LR-TUXP-Ecu-RIOS Zmm-LR-TUXP-Mex-ACAT Zmm-LR-TUXP-Mex-MATA
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/G                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  ?/?
##              ZDP_0902b              ZDP_0828b              ZDP_0419a
## 1 Zmm-LR-TUXP-Mex-URSU Zmm-LR-TUXP-Ven-PORT-1 Zmm-LR-UCHI-Ecu-LOJA-2
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/C                  C/G                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##               ZDP_0070a              ZDP_0145a              ZDP_1164a
## 1 Zmm-LR-UCHU-Bol-CHUQ-1 Zmm-LR-UCHU-Bol-SANT-2 Zmm-LR-UCHU-Per-SAND-3
## 2                  ZMMLR                  ZMMLR                  ZMMLR
## 3                    C/C                    C/C                    C/C
## 4                    A/A                    A/A                    A/A
## 5                    G/T                    G/T                    T/T
## 6                    A/G                    A/G                    G/G
##               ZDP_1037a              ZDP_0996a            ZDP_0654b
## 1 Zmm-LR-VAND-Mex-ESCU-1 Zmm-LR-VAND-Mex-MAPA-2 Zmm-LR-WDEN-usa-AR
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    ?/?                    C/C                  C/C
## 4                    A/A                    A/A                  A/A
## 5                    T/T                    T/T                  ?/?
## 6                    G/G                    G/G                  G/G
##              ZDP_0652b            ZDP_0712b            ZDP_0736b
## 1 Zmm-LR-WDEN-usa-VA-1 Zmm-LR-WHUL-usa-IL-1 Zmm-LR-WINN-usa-NE-2
## 2                ZMMLR                ZMMLR                ZMMLR
## 3                  C/G                  C/G                  C/C
## 4                  A/A                  A/A                  A/A
## 5                  T/T                  T/T                  T/T
## 6                  G/G                  G/G                  G/G
##         ZDP_0668b          ZDP_0657b            ZDP_0099b            ZDP_0374a
## 1 Zmm-LR-WOST-usa Zmm-LR-YANK-usa-SD Zmm-LR-YGNO-Bol-LAPA Zmm-LR-YGNO-Ecu-SANT
## 2            ZMMLR              ZMMLR                ZMMLR                ZMMLR
## 3              C/G                C/G                  C/C                  ?/?
## 4              A/A                A/A                  A/A                  A/A
## 5              T/T                T/T                  T/T                  T/T
```

```
## 6             G/G                G/G                  G/G                  G/G
##         ZDP_0716b              ZDP_1080a              ZDP_0454a
## 1 Zmm-LR-YJEL-usa Zmm-LR-YNQA-Ecu-MORO-3 Zmm-LR-YNQV-Ecu-MORO-2
## 2           ZMMLR                  ZMMLR                  ZMMLR
## 3             C/C                    C/G                    C/C
## 4             A/A                    A/A                    A/A
## 5             T/T                    T/T                    T/T
## 6             G/G                    G/G                    G/G
##              ZDP_0694b              ZDP_0903a            ZDP_0298a
## 1 Zmm-LR-YPRL-usa-IA-1 Zmm-LR-YUCA-Col-ARME-2 Zmm-LR-YUNG-Ecu-MALD
## 2                ZMMLR                  ZMMLR                ZMMLR
## 3                  C/C                    C/C                  C/C
## 4                  A/A                    A/A                  A/A
## 5                  T/T                    T/T                  T/T
## 6                  G/G                    G/G                  G/G
##                ZDP_0599a            ZDP_0550a            ZDP_0244b
## 1 Zmm-LR-YUNG-Ecu-NAPO-3 Zmm-LR-ZAMO-Mex-VIBO Zmm-LR-ZAPC-Mex-COMA
## 2                  ZMMLR                  ZMMLR                ZMMLR
## 3                    C/C                    ?/?                  C/C
## 4                    A/A                    A/A                  A/A
## 5                    T/T                    T/T                  G/T
## 6                    G/G                    ?/?                  A/G
##                ZDP_0569a       ZDP_0243a            ZDP_0246b
## 1 Zmm-LR-ZAPC-Mex-NILT-3 Zmm-LR-ZAPG-Mex Zmm-LR-ZAPG-Mex-ROSE
## 2                  ZMMLR           ZMMLR                ZMMLR
## 3                    C/G             C/C                  G/G
## 4                    A/A             A/A                  A/A
## 5                    T/T             T/T                  T/T
## 6                    G/G             G/G                  G/G
##            ZDP_0737b            BKN_012            BKN_029            BKN_033
## 1 Zmm-LR-ZUNI-usa-ND Zmm-MR01-S4-ARAG_f Zmm-MR03-S5-BOLI_f Zmm-MR06-S5-CHAP_f
## 2              ZMMLR              ZMMMR              ZMMMR              ZMMMR
## 3                C/G                ?/?                C/C                C/C
## 4                A/A                A/A                A/A                A/A
## 5                T/T                G/G                T/T                T/T
## 6                G/G                A/A                G/G                G/G
##              BKN_032            BKN_015            BKN_021            BKN_017
## 1 Zmm-MR09-S5-CRVR_f Zmm-MR12-S6-HAVA_f Zmm-MR16-S6-PEPI_f Zmm-MR19-S6-SDOM_f
## 2              ZMMMR              ZMMMR              ZMMMR              ZMMMR
## 3                C/C                G/G                C/C                C/C
## 4                A/A                A/A                A/A                A/A
## 5                T/T                T/T                T/T                T/T
## 6                G/G                G/G                G/G                G/G
##              BKN_025            BKN_010        S1607       S1643        CU0005
## 1 Zmm-MR22-S5-TUXP_f Zmm-MR25-S4-PORO_f Zmm-IL-38-11 Zmm-IL-A188 Zmm-IL-A272_f
## 2              ZMMMR              ZMMMR        ZMMIL       ZMMIL         ZMMIL
## 3                ?/?                C/C          C/C         C/C           C/C
## 4                A/A                A/A          A/A         A/A           A/A
## 5                T/T                T/T          T/T         T/T           T/T
## 6                G/G                G/G          ?/?         G/G           G/G
##          CU0006       S1646        CU0009       S1647         CU0013
## 1 Zmm-IL-A556_f Zmm-IL-A632 Zmm-IL-A654_f Zmm-IL-A679 Zmm-IL-Ab28A_f
## 2         ZMMIL       ZMMIL         ZMMIL       ZMMIL          ZMMIL
## 3           G/G         G/G           G/G         C/C            C/C
```

```
## 4           A/A         A/A           A/A         A/A            A/A
## 5           T/T         T/T           T/T         T/T            T/T
## 6           G/G         G/G           G/G         G/G            G/G
##         S1642        CU0017     S1651       CU0020       CU0023
## 1 Zmm-IL-B104 Zmm-IL-B115_f Zmm-IL-B2 Zmm-IL-B52_f Zmm-IL-B68_f
## 2       ZMMIL         ZMMIL     ZMMIL        ZMMIL        ZMMIL
## 3         C/C           C/C       C/C          C/C          G/G
## 4         A/A           A/A       A/A          A/A          A/A
## 5         T/T           T/T       T/T          T/T          T/T
## 6         G/G           G/G       G/G          G/G          G/G
##              CU0024       CU0027      S1624       S1653       CU0031
## 1 Zmm-IL-B73Htrhm_f Zmm-IL-B77_f Zmm-IL-B97 Zmm-IL-C123 Zmm-IL-CH9_f
## 2             ZMMIL        ZMMIL      ZMMIL       ZMMIL        ZMMIL
## 3               C/C          C/C        C/C         C/C          G/G
## 4               A/A          A/A        A/A         A/A          A/A
## 5               T/T          T/T        T/T         T/T          T/T
## 6               G/G          G/G        G/G         G/G          G/G
##           CU0034        CU0037         CU0039        S1601         CU0043
## 1 Zmm-IL-CI21E_f Zmm-IL-CI3A_f Zmm-IL-CI90C_f Zmm-IL-CM174 Zmm-IL-CML10_f
## 2          ZMMIL         ZMMIL          ZMMIL        ZMMIL          ZMMIL
## 3            G/G           ?/?            C/C          G/G            C/C
## 4            A/A           A/A            A/A          A/A            A/A
## 5            T/T           T/T            T/T          T/T            T/T
## 6            G/G           G/G            G/G          G/G            G/G
##           CU0045           CU0048          CU0051          SYN072
## 1 Zmm-IL-CML11_f Zmm-IL-CML157Q_f Zmm-IL-CML220_f Zmm-IL-CML247_v
## 2          ZMMIL            ZMMIL           ZMMIL           ZMMIL
## 3            C/C              G/G             C/C             G/G
## 4            A/A              A/A             A/A             A/A
## 5            T/T              G/G             T/T             T/T
## 6            G/G              A/A             G/G             G/G
##            CU0055          CU0057          CU0060          CU0062
## 1 Zmm-IL-CML261_f Zmm-IL-CML281_f Zmm-IL-CML314_f Zmm-IL-CML323_f
## 2           ZMMIL            ZMMIL           ZMMIL           ZMMIL
## 3             C/C              G/G             G/G             C/C
## 4             A/A              A/A             A/A             A/A
## 5             T/T              T/T             T/T             T/T
## 6             G/G              G/G             G/G             G/G
##            CU0065         CU0067         SYN094         CU0071       S1637
## 1 Zmm-IL-CML332_f Zmm-IL-CML38_f Zmm-IL-CML52_v Zmm-IL-CML77_f Zmm-IL-CMV3
## 2           ZMMIL          ZMMIL          ZMMIL          ZMMIL       ZMMIL
## 3             G/G            ?/?            C/C            C/C         C/C
## 4             A/A            A/A            A/A            A/A         A/A
## 5             G/G            T/T            T/T            T/T         T/T
## 6             A/A            G/G            G/G            G/G         G/G
##           CU0076        CU0079          CU0082      S1621         CU0087
## 1 Zmm-IL-CO255_f Zmm-IL-DE_3_f Zmm-IL-E2558W_f Zmm-IL-F44 Zmm-IL-GA209_f
## 2          ZMMIL         ZMMIL           ZMMIL      ZMMIL          ZMMIL
## 3            C/C           ?/?             C/C        C/C            G/G
## 4            A/A           A/A             A/A        A/A            A/A
## 5            T/T           T/T             T/T        T/T            T/T
## 6            G/G           G/G             G/G        G/G            G/G
##         CU0089      S1610        S1616         S1626         S1661
## 1 Zmm-IL-H49_f Zmm-IL-H95 Zmm-IL-HP301 Zmm-IL-I137TN Zmm-IL-IA2132
```

```
## 2        ZMMIL      ZMMIL        ZMMIL         ZMMIL         ZMMIL
## 3          G/G        C/C          C/C           G/G           C/C
## 4          A/A        A/A          A/A           A/A           A/A
## 5          T/T        T/T          T/T           G/G           T/T
## 6          G/G        G/G          G/G           A/A           G/G
##           CU0096          S0795          CU0098      S1639        CU0102
## 1 Zmm-IL-IDS69_f Zmm-IL-IL14H_a Zmm-IL-Il677a_f Zmm-IL-K55 Zmm-IL-Ki14_f
## 2          ZMMIL          ZMMIL           ZMMIL      ZMMIL         ZMMIL
## 3            C/C            C/C             C/C        C/C           C/C
## 4            A/A            A/A             A/A        A/A           A/A
## 5            T/T            T/T             T/T        T/T           T/T
## 6            G/G            G/G             G/G        G/G           G/G
##         SYN150        SYN153        CU0108         SYN162       S1627
## 1 Zmm-IL-Ki3_v Zmm-IL-Ky21_v Zmm-IL-L317_f Zmm-IL-M162W_v Zmm-IL-Mo17
## 2        ZMMIL         ZMMIL         ZMMIL          ZMMIL       ZMMIL
## 3          C/C           C/C           ?/?            C/C         C/C
## 4          A/A           A/A           A/A            A/A         A/A
## 5          T/T           T/T           T/T            T/T         T/T
## 6          G/G           G/G           G/G            G/G         G/G
##          CU0112        CU0115       CU0118         CU0121       S1631
## 1 Zmm-IL-MO1W_f Zmm-IL-Mo45_f Zmm-IL-MoG_f Zmm-IL-MS153_f Zmm-IL-N192
## 2         ZMMIL         ZMMIL        ZMMIL          ZMMIL       ZMMIL
## 3           G/G           C/C          G/G            C/C         C/C
## 4           A/A           A/A          A/A            A/A         A/A
## 5           T/T           T/T          T/T            T/T         T/T
## 6           G/G           G/G          G/G            G/G         G/G
##         CU0125         CU0128        S1658         CU0132         CU0135
## 1 Zmm-IL-N7A_f Zmm-IL-NC232_f Zmm-IL-NC250 Zmm-IL-NC262_f Zmm-IL-NC294_f
## 2        ZMMIL          ZMMIL        ZMMIL          ZMMIL          ZMMIL
## 3          G/G            G/G          G/G            C/C            C/C
## 4          A/A            A/A          A/A            A/A            A/A
## 5          T/T            T/T          T/T            T/T            T/T
## 6          G/G            G/G          G/G            G/G            G/G
##           CU0138         CU0140         CU0143         CU0146        CU0149
## 1 Zmm-IL-NC298_f Zmm-IL-NC304_f Zmm-IL-NC314_f Zmm-IL-NC324_f Zmm-IL-NC33_f
## 2          ZMMIL          ZMMIL          ZMMIL          ZMMIL         ZMMIL
## 3            C/C            C/C            ?/?            C/C           G/G
## 4            A/A            A/A            A/A            A/A           A/A
## 5            T/T            T/T            T/T            T/T           T/T
## 6            G/G            G/G            G/G            G/G           G/G
##           CU0152         CU0155        S1629         SYN199         CU0161
## 1 Zmm-IL-NC340_f Zmm-IL-NC346_f Zmm-IL-NC352 Zmm-IL-NC358_v Zmm-IL-NC364_f
## 2          ZMMIL          ZMMIL        ZMMIL          ZMMIL          ZMMIL
## 3            G/G            C/C          C/C            C/C            C/C
## 4            A/A            A/A          A/A            A/A            A/A
## 5            ?/?            T/T          T/T            T/T            T/T
## 6            G/G            G/G          G/G            G/G            G/G
##           CU0164        SYN209       S1659      S1660         CU0170
## 1 Zmm-IL-ND246_f Zmm-IL-Oh43_v Zmm-IL-Oh7B Zmm-IL-P39 Zmm-IL-Pa875_f
## 2          ZMMIL         ZMMIL       ZMMIL      ZMMIL          ZMMIL
## 3            C/C           C/C         C/C        C/C            C/C
## 4            A/A           A/A         A/A        A/A            A/A
## 5            T/T           T/T         T/T        T/T            T/T
## 6            G/G           G/G         G/G        G/G            G/G
```

```
##           CU0173        CU0176          CU0178        CU0180        CU0182
## 1 Zmm-IL-R109B_f Zmm-IL-R229_f Zmm-IL-SC213R_f Zmm-IL-SD44_f Zmm-IL-T232_f
## 2          ZMMIL         ZMMIL           ZMMIL         ZMMIL         ZMMIL
## 3            C/C           C/C             C/C           C/C           C/C
## 4            A/A           A/A             A/A           A/A           A/A
## 5            T/T           T/T             T/T           T/T           T/T
## 6            G/G           G/G             G/G           G/G           G/G
##           SYN240         CU0185         CU0188         CU0190        CU0193
## 1 Zmm-IL-Tx303_v Zmm-IL-Tzi11_f Zmm-IL-Tzi25_f Zmm-IL-U267Y_f Zmm-IL-Va17_f
## 2          ZMMIL          ZMMIL          ZMMIL          ZMMIL         ZMMIL
## 3            C/C            C/C            C/C            C/C           G/G
## 4            A/A            A/A            A/A            A/A           A/A
## 5            T/T            T/T            G/G            T/T           T/T
## 6            G/G            G/G            A/A            G/G           G/G
##          CU0195        CU0198        S0351                CU0200          S1636
## 1 Zmm-IL-Va35_f Zmm-IL-Va99_f Zmm-IL-W153R Zmm-IL-W22_R-r:std_f Zmm-IL-W64A_b_s
## 2         ZMMIL         ZMMIL        ZMMIL                 ZMMIL           ZMMIL
## 3           C/C           C/C          C/C                   ?/?             G/G
## 4           A/A           A/A          A/A                   A/A             A/A
## 5           T/T           T/T          T/T                   T/T             T/T
## 6           G/G           G/G          G/G                   G/G             G/G
##              CU0202   SNP_ID
## 1 Zmm-IL-Yu796_NS_f   JG_OTU
## 2             ZMMIL    Group
## 3               C/C abph1.20
## 4               A/A abph1.22
## 5               T/T    ae1.3
## 6               G/G    ae1.4
```

```
### teosinte genotypes Same logic as maize above
teosinte_genotypes_transposed <- t(teosinte_genotypes)
teosinte_genotypes_transposed <- as.data.frame(teosinte_genotypes_transposed)
colnames(teosinte_genotypes_transposed) <- teosinte_genotypes_transposed[1, ]
teosinte_genotypes_transposed <- teosinte_genotypes_transposed[-1, ]
teosinte_genotypes_transposed$SNP_ID <- rownames(teosinte_genotypes_transposed)
rownames(teosinte_genotypes_transposed) <- NULL
```

```
# Merging

## Select only the essential mapping columns (ID, Chr, Pos)
snp_pos_data_needed_columns <- snp_pos_data %>% select(SNP_ID, Chromosome, Position)
```

```
# joining genotype data and snp position data

## maize genotypes and snp position (i used `dim` to qc the files to make sure they are properly joined
merged_maize_geno_snp <- inner_join(snp_pos_data_needed_columns, maize_genotypes_transposed, by = "SNP_
dim(merged_maize_geno_snp)
```

```
## [1] 983 528
```

```
dim(snp_pos_data_needed_columns)
```

```
## [1] 983   3
```

```r
dim(maize_genotypes_transposed)
```

```
## [1] 985 526
```

```r
## teosinte genotypes and snp position
merged_teosinte_geno_snp <- inner_join(snp_pos_data_needed_columns, teosinte_genotypes_transposed, by =
dim(merged_teosinte_geno_snp)
```

```
## [1] 983 328
```

```r
dim(snp_pos_data_needed_columns)
```

```
## [1] 983   3
```

```r
dim(teosinte_genotypes_transposed)
```

```
## [1] 985 326
```

```r
# maize: 10 files (1 for each chromosome) with SNPs ordered based on increasing position values and wit

## Filter for chromosomes 1 through 10
maize_subset <- merged_maize_geno_snp[merged_maize_geno_snp$Chromosome %in% 1:10, ]

# Split the data frame into a LIST of 10 data frames (one per chromosome)
maize_list <- split(maize_subset, maize_subset$Chromosome)

# Define the  function, sort and define filename for saving
process_and_save_per_chr_increase <- function(chr) {
  df <- maize_list[[chr]]
  df_sorted <- df[order(as.numeric(df$Position)), ]
  file_name <- paste0("maize_chr", chr, "_increasing.txt")
  write.table(df_sorted, file = `file_name`, sep = "\t", row.names = FALSE, quote = FALSE)
}
# use Apply to run through all chromosome lists
lapply(names(maize_list), process_and_save_per_chr_increase)
```

```
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion

## [[1]]
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```

```
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## [[4]]
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## [[9]]
## NULL
##
## [[10]]
## NULL
```

```r
# maize: 10 files (1 for each chromosome) with SNPs ordered based on decreasing position values and wit

process_and_save_per_chr_decrease <- function(chr) {
  df <- maize_list[[chr]]
   # Replace "??" with "-" across the entire data frame
  df[df == "?/?"] <- "-/-"
  df_sorted <- df[order(as.numeric(df$Position), decreasing = TRUE), ]
  file_name <- paste0("maize_chr", chr, "_decreasing.txt")
  write.table(df_sorted, file = `file_name`, sep = "\t", row.names = FALSE, quote = FALSE)
}
# use Apply to run through all chromosome lists
lapply(names(maize_list), process_and_save_per_chr_decrease)
```

```
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
```

```
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## NULL
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## [[9]]
## NULL
##
## [[10]]
## NULL
```

```r
# teosinte (same logic as above)
# teosinte: 10 files (1 for each chromosome) with SNPs ordered based on increasing position values and

## Filter for chromosomes 1 through 10
teosinte_subset <- merged_teosinte_geno_snp[merged_teosinte_geno_snp$Chromosome %in% 1:10, ]

# Split the data frame into a LIST of 10 data frames (one per chromosome)
teosinte_list <- split(teosinte_subset, teosinte_subset$Chromosome)
process_and_save_per_chr_increase <- function(chr) {
  df <- teosinte_list[[chr]]
  df_sorted <- df[order(as.numeric(df$Position)), ]
  file_name <- paste0("teosinte_chr", chr, "_increasing.txt")
  write.table(df_sorted, file = `file_name`, sep = "\t", row.names = FALSE, quote = FALSE)
}
# use Apply to run through all chromosome lists
lapply(names(teosinte_list), process_and_save_per_chr_increase)
```

```
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
## Warning in order(as.numeric(df$Position)): NAs introduced by coercion
```

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## [[8]]
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## [[9]]
## NULL
##
## [[10]]
## NULL
```

```r
# teosinte files (1 for each chromosome) with SNPs ordered based on decreasing position values and with

process_and_save_per_chr_decrease <- function(chr) {
  df <- teosinte_list[[chr]]
   # Replace "??" with "-" across the entire data frame
  df[df == "?/?"] <- "-/-"
  df_sorted <- df[order(as.numeric(df$Position), decreasing = TRUE), ]
  file_name <- paste0("teosinte_chr", chr, "_decreasing.txt")
  write.table(df_sorted, file = `file_name`, sep = "\t", row.names = FALSE, quote = FALSE)
}
# use Apply to run through all chromosome lists
lapply(names(teosinte_list), process_and_save_per_chr_decrease)
```

```
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
## Warning in order(as.numeric(df$Position), decreasing = TRUE): NAs introduced by
## coercion
```

```
## [[1]]
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```

```
## [[3]]
## NULL
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##
## [[10]]
## NULL
```
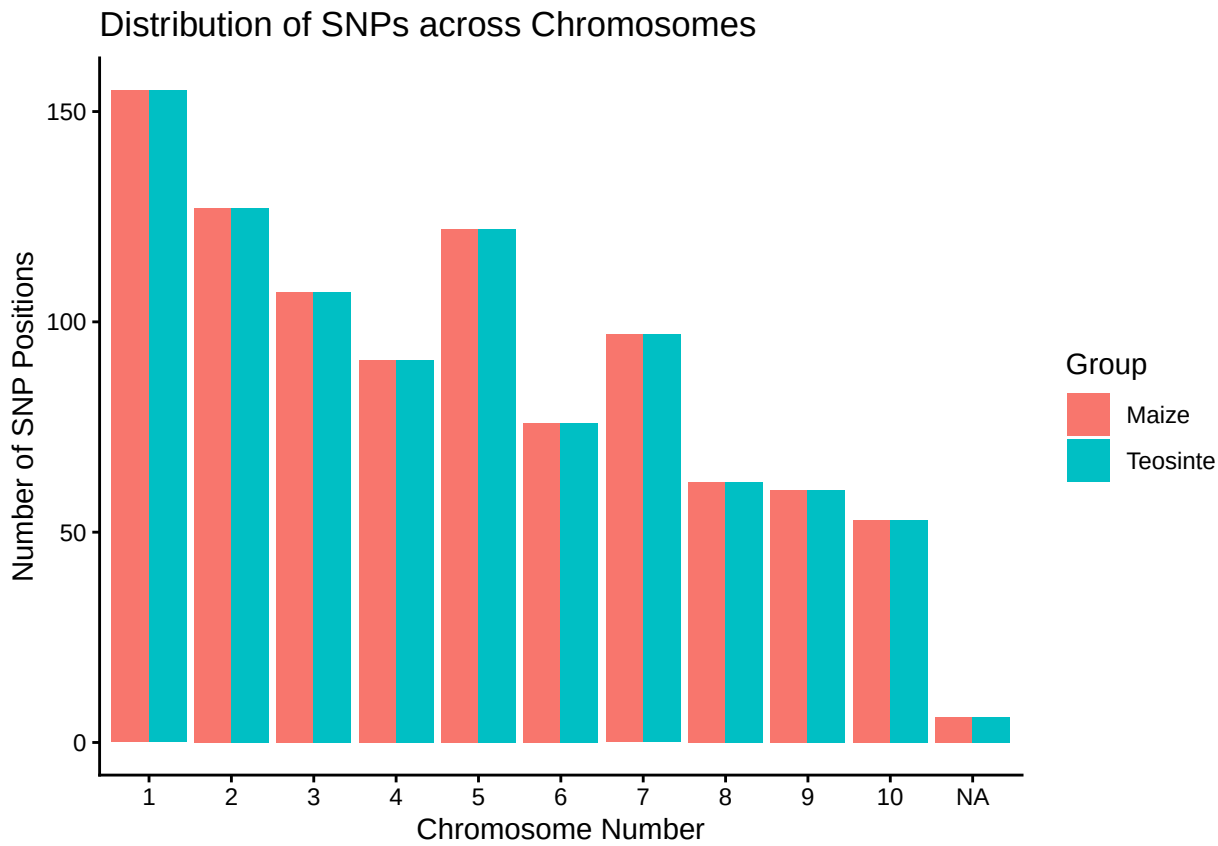
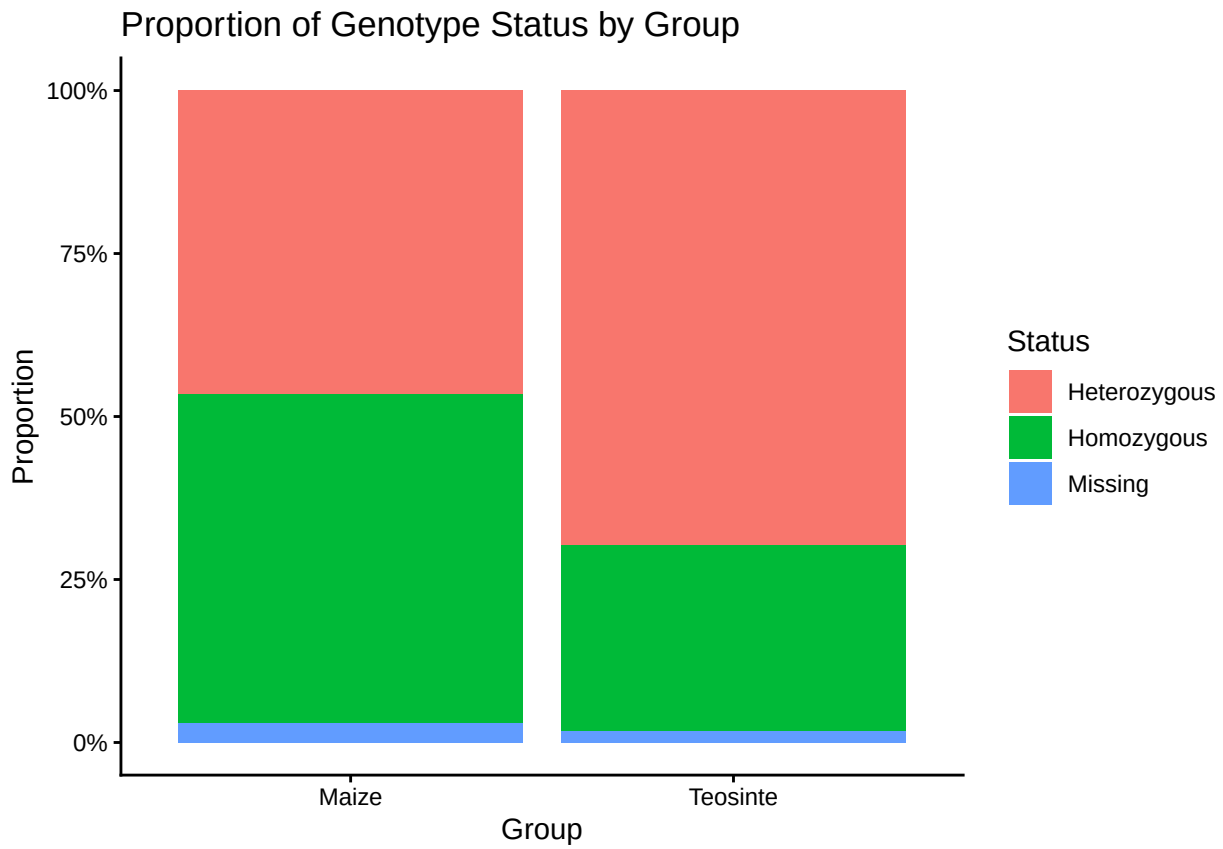

```r
# Visuaization

## Prepare unified dataset for plotting
maize_data <- merged_maize_geno_snp %>% mutate(Group = "Maize")
teosinte_data <- merged_teosinte_geno_snp %>% mutate(Group = "Teosinte")

### i mae these column the same type of variable to facilitate with combining the dataset
maize_data <- maize_data %>%
  mutate(across(c(SNP_ID, Chromosome, Position), as.character), Group = "Maize")

teosinte_data <- teosinte_data %>%
  mutate(across(c(SNP_ID, Chromosome, Position), as.character), Group = "Teosinte")

# Combine datasets
combined_data <- bind_rows(maize_data, teosinte_data)

# Remove rows with unknown chromosome/position
combined_data <- combined_data %>%
  filter(!is.na(Chromosome) & Chromosome != "unknown") %>%
  mutate(Chromosome = factor(Chromosome, levels = c(1:10)))

# Visualization 1: SNPs per Chromosome

## Question: What is the distribution of SNPs on and across chromosomes? ##Are there more SNP positions

ggplot(combined_data, aes(x = Chromosome, fill = Group)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of SNPs across Chromosomes",
       x = "Chromosome Number",
       y = "Number of SNP Positions",
```

```
       fill = "Group") +
  theme_classic()
```

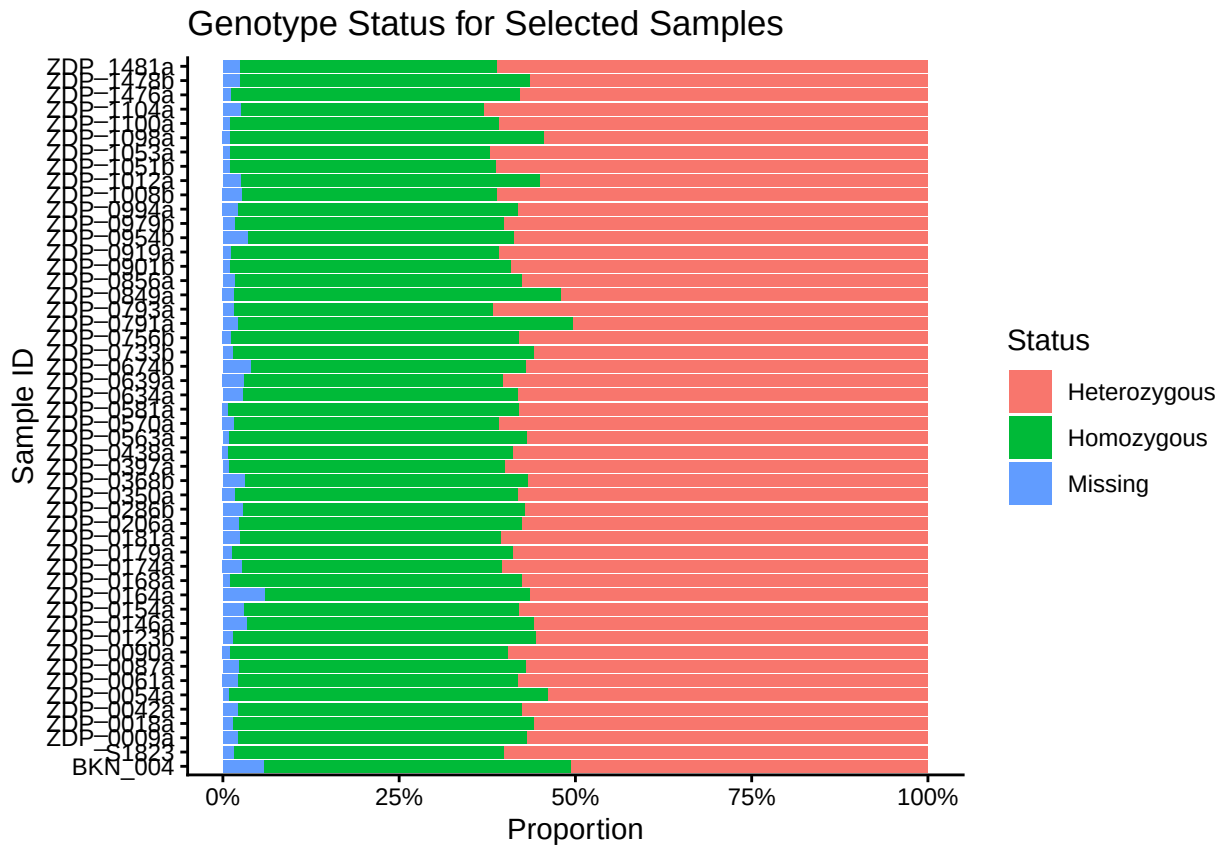



```
# Visualization 2: Missing Data and Heterozygosity
# Question: What is the proportion of homozygous, heterozygous, and missing data?

# Reshape data to long format for analysis
# I exclude the metadata columns to focus on sample genotypes
genotype_data <- combined_data %>%
  select( -SNP_ID, -Chromosome, -Position) %>%
  pivot_longer(cols = -Group, names_to = "Sample", values_to = "Genotype")

# Define genotype categories
genotype_summary <- genotype_data %>%
  mutate(Status = case_when(
    Genotype == "?/?" ~ "Missing",
    substr(Genotype, 1, 1) == substr(Genotype, 3, 3) ~ "Homozygous",
    TRUE ~ "Heterozygous"
  ))

# Plotting the proportions normalized to 100%
ggplot(genotype_summary, aes(x = Group, fill = Status)) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of Genotype Status by Group",
       x = "Group",
```

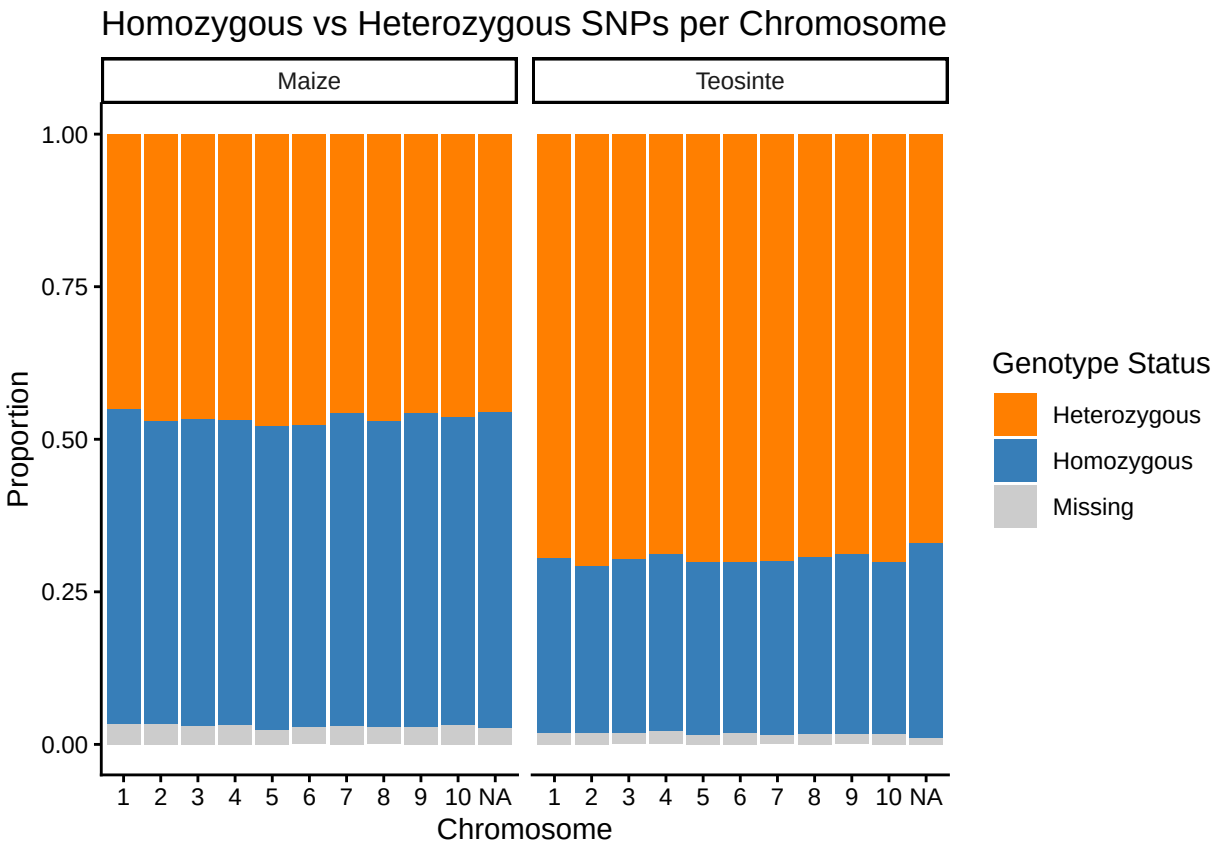
```
       y = "Proportion",
       fill = "Status") +
  scale_y_continuous(labels = scales::percent) +
  theme_classic()
```




```
# visualization 3
## Plot by individual samples (top 50 for readability)
sample_subset <- genotype_summary %>%
  filter(Sample %in% unique(genotype_summary$Sample)[1:50])

ggplot(sample_subset, aes(x = Sample, fill = Status)) +
  geom_bar(position = "fill") +
  coord_flip() +
  labs(title = "Genotype Status for Selected Samples",
       x = "Sample ID",
       y = "Proportion",
       fill = "Status") +
  scale_y_continuous(labels = scales::percent) +
  theme_classic()
```




```r
# visualization 4
## Detailed view of Homozygous, Heterozygous, and Missing data per chromosome, side-by-side for Maize a
combined_long <- combined_data %>%
  pivot_longer(cols = -c(SNP_ID, Chromosome, Position, Group),
               names_to = "Sample",
               values_to = "Genotype") %>%
  mutate(Status = case_when(
    Genotype == "?/?" ~ "Missing",
    substr(Genotype, 1, 1) == substr(Genotype, 3, 3) ~ "Homozygous",
    TRUE ~ "Heterozygous"
  ))

# 2. Plot using the new 'combined_long' object
ggplot(combined_long, aes(x = factor(Chromosome), fill = Status)) +
  geom_bar(position = "fill") +
  facet_grid(~Group) +
  scale_fill_manual(values = c("Homozygous" = "#377eb8",
                               "Heterozygous" = "#ff7f00",
                               "Missing" = "grey80")) +
  theme_classic() +
  labs(title = "Homozygous vs Heterozygous SNPs per Chromosome",
       x = "Chromosome",
       y = "Proportion",
       fill = "Genotype Status")
```


Homozygous vs Heterozygous SNPs per Chromosome
Maize
Teosinte
1.00
0.75
0.50
0.25
0.00
Proportion
1 2 3 4 5 6 7 8 9 10 NA
1 2 3 4 5 6 7 8 9 10 NA
Chromosome
Genotype Status
Heterozygous
Homozygous
Missing